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MODULAR ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM

Abstract

A head unit system for controlling an object includes a head unit device that include shear thickening fluid (STF) and a chamber configured to contain the STF. The chamber further includes a set of gates between a front channel and a back channel. The set of gates includes a bypass opening set. The head unit device further includes a piston housed at least partially radially within the chamber. The set of gates is configured to control flow of the STF between the front channel and the back channel to control rotational movement of the object. An accessory module assists in control of the object.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present U.S. Utility patent application claims priority pursuant to 35 U.S.C. § 120 as a continuation of U.S. Utility application Ser. No. 18/646,943, entitled “MODULAR ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM,” filed Apr. 26, 2024, issuing Apr. 29, 2025 as U.S. Pat. No. 12,286,060, which claims priority pursuant to 35 U.S.C. § 120 as a continuation-in-part of U.S. Utility application Ser. No. 18/586,061, entitled “MODULAR ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM,” filed Feb. 23, 2024, issuing Apr. 29, 2025 as U.S. Pat. No. 12,286,829, which claims priority pursuant to 35 U.S.C. § 120 as a continuation-in-part of U.S. Utility application Ser. No. 18/515,620, entitled “MODULAR ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM,” filed Nov. 21, 2023, issuing Apr. 29, 2025 as U.S. Pat. No. 12,287,024, which claims priority pursuant to 35 U.S.C. § 120 as a continuation-in-part of U.S. Utility application Ser. No. 18/071,680, entitled “ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM,” filed Nov. 30, 2022, issued Nov. 28, 2023 as U.S. Pat. No. 11,828,309, which claims priority pursuant to 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/427,949, entitled “SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM”, filed Nov. 25, 2022, expired, all of which are hereby incorporated herein by reference in their entirety and made part of the present U.S. Utility patent application for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT
[0002] Not Applicable.
INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC
[0003] Not Applicable.
BACKGROUND OF THE INVENTION
Technical Field of the Invention
[0004] This invention relates generally to systems that measure and control mechanical movement and more particularly to sensing and controlling of a linear and/or rotary movement mechanism that includes a chamber with dilatant fluid (e.g., a shear thickening fluid).
Description of Related Art
[0005] Many mechanical mechanisms are subject to undesired movement that can lead to annoying sounds, property damage and/or loss, and personal injury and even death. Desired and undesired

movements of the mechanical mechanisms may involve a wide range of forces. A need exists to control the wide range of forces to solve these problems.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0006] FIG. **1**A is a schematic block diagram of an embodiment of a mechanical and computing system in accordance with the present invention;

[0007] FIG. **1**B is a graph of viscosity vs. shear rate for an aspect of an embodiment of a mechanical and computing system in accordance with the present invention;

[0008] FIG. **1**C is a graph of plunger velocity vs. force applied to the plunger for an aspect of an embodiment of a mechanical and computing system in accordance with the present invention;

[0009] FIG. **2**A is a schematic block diagram of an embodiment of a computing entity of a computing system in accordance with the present invention;

[0010] FIG. **2**B is a schematic block diagram of an embodiment of a computing device of a computing system in accordance with the present invention;

[0011] FIG. **3** is a schematic block diagram of another embodiment of a computing device of a computing system in accordance with the present invention;

[0012] FIG. **4** is a schematic block diagram of an embodiment of an environment sensor module of a computing system in accordance with the present invention;

[0013] FIGS. **5**A-**5**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0014] FIGS. **6**A-**6**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0015] FIGS. **7**A-**7**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0016] FIGS. **8**A-**8**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0017] FIGS. **9**A-**9**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0018] FIGS. **10**A-**10**D are schematic block diagrams of an embodiment of a mechanical system to control an object in accordance with the present invention;

[0019] FIGS. **11**A-**11**D are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention;

[0020] FIGS. **12**A-**12**F are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention;

[0021] FIGS. **13**A-**13**D are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention;

[0022] FIGS. **14**A-**14**C are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention; and

[0023] FIGS. **15**A-**15**D are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0024] FIG. **1**A is a schematic block diagram of an embodiment of a mechanical and computing system that includes a set of head units **10-1** through **10**-N, objects **12-1** through **12-3**, computing entities **20-1** through **20**-N associated with the head units **10-1** through **10**-N, and a computing entity **22**. The objects include any object that has mass and moves. Examples of an object include a door, an aircraft wing, a portion of a building support mechanism, and a particular drivetrain, etc.

[0025] The cross-sectional view of FIG. **1**A illustrates a head unit that includes a chamber **16**, a piston **36**, a plunger **28**, a plunger bushing **32**, and a chamber bypass **40**. The chamber **16** contains a shear thickening fluid (STF) **42**. The chamber **16** includes a back channel **24** and a front channel **26**, where the piston partitions the back channel **24** and the front channel **26**. The piston **36** travels axially within the chamber **16**. The chamber **16** may be a cylinder or any other shape that enables movement of the piston **36** and compression of the STF **42**. The STF **42** is discussed in greater detail with reference to FIGS. **1**B and **1**C.

[0026] The plunger bushing **32** guides the plunger **28** into the chamber **16** in response to force from the object **12-1**. The plunger bushing **32** facilitates containment of the STF within the chamber **16**. The plunger bushing **32** remains in a fixed position relative to the chamber **16** when the force from the object moves the piston **36** within the chamber **16**. In an embodiment the plunger bushing **32** includes an O-ring between the plunger bushing **32** and the chamber **16**. In another embodiment the plunger bushing **32** includes an O-ring between the plunger bushing **32** and the plunger **28**.

[0027] The piston **36** includes a piston bypass **38** between opposite sides of the piston to facilitate flow of a portion of the STF between the opposite sides of the piston (e.g., between the back channel **24** and the front channel **26**) when the piston travels through the chamber in an inward or an outward direction.

[0028] Alternatively, or in addition to, the chamber bypass **40** is configured between opposite ends of the chamber **16**, wherein the chamber bypass **40** facilitates flow of a portion of the STF between the opposite ends of the chamber when the piston travels through the chamber in the inward or outward direction (e.g., between the back channel **24** and the front channel **26**).

[0029] In alternative embodiments, the piston bypass **38** and the chamber bypass **40** includes mechanisms to enable STF flow in one direction and not an opposite direction. In further alternative embodiments, a control valve within the piston bypass **38** and/or the chamber bypass **40** controls the STF flow between the back channel **24** and the front channel **26**. Each bypass includes one or more of a one-way check valve and a variable flow valve.

[0030] The plunger **28** is operably coupled to a corresponding object by one of a variety of approaches. A first approach includes a direct connection of the plunger **28** to the object **12-1** such that linear motion in any direction couples from the object **12-1** to the plunger **28**. A second approach includes the plunger **28** coupled to a cap **44** which receives a one way force from a strike **48** attached to the object **12-2**. A third approach includes a pushcap **46** that receives a force from a rotary-to-linear motion conversion component that is attached to the object **12-3**. In an example, the object **12-3** is connected to a camshaft **110** which turns a cam **109** to strike the pushcap **46**.

[0031] In an embodiment, two or more of the head units are coupled by a head unit connector **112**. When so connected, actuation of a piston in a first head unit is essentially replicated in a piston of a second head unit. The head unit connector **112** includes a mechanical element between plungers of the two or more head units and/or direct connection of two or more plungers to a common object. For example, plunger **28** of head unit **10-1** and plunger **28** of head unit **10-2** are directly connected to object **12-1** when utilizing a direct connection.

[0032] Further associated with each head unit is a set of emitters and a set of sensors. For example, head unit **10**-N includes a set of emitters **114**-N−1 through **114**-N-M and a set of sensors **116**-N−1 through **116**-N-M. Emitters includes any type of energy and or field emitting device to affect the STF, either directly or indirectly via other nanoparticles suspended in the STF. Examples of emitter categories include light, audio, electric field, magnetic field, wireless field, etc. Specific examples of fluid manipulation emitters include a variable flow valve associated with a bypass or injector or similar, a mechanical vibration generator, an image generator, a light emitter, an audio transducer, a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter. Specific examples of magnetic field emitters include a Helmholtz coil, a Maxwell coil, a permanent magnet, a solenoid, a superconducting electromagnet, and a radio frequency transmitting coil.

[0033] Sensors include any type of energy and/or field sensing device to output a signal that represents a reaction, motion or position of the STF. Examples of sensor categories include bypass valve position, mechanical position, image, light, audio, electric field, magnetic field, wireless field, etc. Specific examples of fluid flow sensors include a valve opening detector associated with the chamber **16** or any type of bypass (e.g., piston bypass **38**, chamber bypass **40**, a reservoir injector, or similar), a mechanical position sensor, an image sensor, a light sensor, an audio sensor, a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor. Specific examples of magnetic field sensors include a Hall effect sensor, a magnetic coil, a rotating coil magnetometer, an inductive pickup coil, an optical magnetometry sensor, a nuclear magnetic resonance sensor, and a caesium vapor magnetometer.

[0034] The computing entities **20-1** through **20-**N are discussed in detail with reference to FIG. **2**A. The computing entity **22** includes a control module **30** and a chamber database **34** to facilitate storage of history of operation, desired operations, and other aspects of the system.

[0035] In an example of operation, the head unit **10-1** controls motion of the object **12-1** and includes the chamber **16** filled at least in part with the shear thickening fluid **42**, the piston **36** housed at least partially radially within the chamber **16**, and the piston **36** is configured to exert pressure against the shear thickening fluid **42** in response to movement of the piston **36** from a force applied to the piston from the object **12-1**. The movement of the piston **36** includes one of traveling through the chamber **16** in an inward direction or traveling through the chamber **16** in an outward direction. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates.

[0036] The shear thickening fluid **42** (e.g., dilatant non-Newtonian fluid) has nanoparticles of a specific dimension that are mixed in a carrier fluid or solvent. Force applied to the shear thickening fluid **42** results in these nanoparticles stacking up, thus stiffening and acting more like a solid than a flowable liquid when a shear threshold is reached. In particular, viscosity of the shear thickening fluid **42** rises significantly when shear rate is increased to a point of the shear threshold. The relationship between viscosity and shear rates is discussed in greater detail with reference to FIGS. **1**A and **1**B.

[0037] In another example of operation, the object **12-1** applies an inward motion force on the plunger **28** which moves the piston **36** in words within the chamber **16**. As the piston moves inward, shear rate of the shear thickening fluid **42** changes. A sensor **116-1-1** associated with the chamber **16** of the head unit **10-1** outputs chamber I/O **160** to the computing entity **20-1**, where the chamber I/O **160** includes a movement data associated with the STF **42** as a result of the piston **36** moving inwards. Having received the chamber I/O **160**, the computing entity **20-1** interprets the chamber I/O **160** to reproduce the movement data.

[0038] The computing entity **20-1** outputs the movement data as a system message **162** to the computing entity **22**. The control module **30** stores the movement data in the chamber database **34** and interprets the movement data to determine whether to dynamically adjust the viscosity of the shear thickening fluid. Dynamic adjustment of the viscosity results in dynamic control of the movement of the piston **36**, the plunger **28**, and ultimately the object **12-1**. Adjustment of the viscosity affects velocity, acceleration, and position of the piston **36**.

[0039] The control module **30** determines whether to adjust the viscosity based on one or more desired controls of the object **12-1**. The desired controls include accelerating, deaccelerating, abruptly stopping, continuing on a current trajectory, continuing at a constant velocity, or any other movement control. For example, the control module **30** determines to abruptly stop the movement of the object **12-1** when the object **12-1** is a door and the door is detected to be closing at a rate above a maximum closing rate threshold level and when the expected shear rate versus viscosity of the shear thickening fluid **42** requires modification (e.g., boost the viscosity now to slow the door from closing too quickly).

[0040] When determining to modify the viscosity, the control module **30** outputs a system message **162** to the computing entity **20-1**, where the system message **162** includes instructions to immediately boost the viscosity beyond the expected shear rate versus viscosity of the shear thickening fluid **42**. Alternatively, the system message **162** includes specific information on the relationship of viscosity versus shear rate.

[0041] Having received the system message **162**, the computing entity **20-1** determines a set of adjustments to make with regards to the shear thickening fluid **42** within the chamber **16**. The set of adjustments includes one or more of adjusting STF **42** flow through the chamber bypass **40**, adjusting STF **42** flow through the piston bypass **38**, and activating an emitter of a set of emitters **114-1-1** through **114**-N−1. The flow adjustments include regulating within a flow range, stopping, starting, and allowing in one particular direction. For example, the computing entity **20-1** determines to activate emitter **114-1-1** to produce a magnetic field such as to interact with magnetic nanoparticles within the STF **42** to raise the viscosity. The computing entity **20-1** issues another chamber I/O **160** to the emitter **114-1-1** to initiate a magnetic influence process to boost the viscosity of the STF **42**.

[0042] In an alternative embodiment, the computing entity **22** issues another system message **162** to two or more computing entities (e.g., **20-1** and **20-2**) to boost the viscosity for corresponding head units **10-1** and **10-2** when the head unit connector **112** connects head units **10-1** and **10-2** and both head units are controlling the motion of the object **12-1**. For instance, one of the head units informs the computing entity **22** that the object **12-1** is moving too quickly inward and the predicted stopping power of the expected viscosity versus shear rate of the STF **42** of the head unit, even when boosted, will not be enough to slow the object **12-1** to a desired velocity or position. When informed that one head unit, even with a modified viscosity, is not enough to control the object **12-1**, the control module **30** determines how many other head units (e.g., connected via the head unit connector **112**) to apply and to dynamically modify the viscosity.

[0043] In yet another alternative embodiment, the computing entity **22** issues a series of system messages **162** to a set of computing entities associated with a corresponding set of head units to produce a cascading effect of altering of the viscosity of the STF **42** of each of the chambers **16** associated with the set of head units. For example, 3 head units are controlled by 3 corresponding computing entities to adjust viscosity in a time cascaded manner. For instance, head unit **10-1** abruptly changes the viscosity to attempt to slow the object **12-1** followed seconds later by head unit **10-2** abruptly changing the viscosity to attempt to further slow the object **12-1**, followed seconds later by head unit **10-1** abruptly changing the viscosity to attempt to further slow the object **12-1**.

[0044] In a still further alternative embodiment, the computing entity **22** conditionally issues each message of the series of system messages **162** to the set of computing entities associated with the corresponding set of head units to produce the cascading effect of altering of the viscosity of the STF **42** of each of the chambers **16** associated with the set of head units only when a most recent adaptation of viscosity is not enough to slow the object **12-1** with desired results. For example, the 3 head units are controlled by the 3 corresponding computing entities to adjust viscosity in a conditional time cascaded manner. For instance, head unit **10-1** abruptly changes the viscosity to attempt to slow the object **12-1** followed seconds later by head unit **10-2** abruptly changing the viscosity if head unit **10-1** was unsuccessful to attempt to further slow the object **12-1**, followed seconds later by head unit **10-2** abruptly changing the viscosity if head unit **10-2** was unsuccessful to attempt to further slow the object **12-1**.

[0045] FIG. **1**B is a graph of viscosity vs. shear rate for an aspect of an embodiment of a mechanical and computing system that includes a chamber, a shear thickening fluid, and a piston that moves through the chamber applying forces on the shear thickening fluid. The shear thickening fluid includes a non-Newtonian fluid since the relationship between shear rate and viscosity is nonlinear.

[0046] A relationship between compressive impulse (e.g., shear rate) and the viscosity of the shear thickening fluid is nonlinear and may comprise one or more inflection points as the piston travels within the chamber in response to different magnitudes of forces and different accelerations. The viscosity of the STF may also be a function of other influences, such as electric fields, acoustical waves, magnetic fields, and other similar influences. As a first example of a response of a shear thickening fluid, a first range of shear rates in zone A has a decreasing viscosity as the shear rate increases and then in a second range of shear rates in zone B the viscosity increases abruptly. As a second example of a response of a diluted shear thickening fluid, the first range of shear rates in zone A extends to a higher level of shear rates with the decreasing viscosity and then in the still higher second range of shear rates in zone B the viscosity increases abruptly similar to that of the shear thickening include.

[0047] The shear thickening fluid includes particles within a solvent. Examples of particles of the shear thickening fluid include oxides, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, or a mixture thereof. Further examples of the particles of the shear thickening fluid include SiO2, polystyrene, or polymethylmethacrylate.

[0048] The particles are suspended in a solvent. Example components of the solvent include water, a salt, a surfactant, and a polymer. Further example components of the solvent include ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone or a mixture thereof. Example particle diameters range from less than 100 μm to less than 1 millimeter. In an instance, the shear thickening fluid is made of silica particles suspended in polyethylene glycol at a volume fraction of approximately 0.57 with the silica particles having an average particle diameter of approximately 446 nm. As a result, the shear thickening fluid exhibits a shear thickening transition at a shear rate of approximately 102-103 s-1.

[0049] A volume fraction of particles dispersed within the solvent distinguishes the viscosity versus shear rate of different shear thickening fluids. The viscosity of the STF changes in response to the applied shear stress. At rest and under weak applied shear stress, a STF may have a fairly constant or even slightly decreasing viscosity because the random distribution of particles causes the particles to frequently collide. However, as a greater shear stress is applied so that the shear rate increases, the particles flow in a more streamlined manner. However, as an even greater shear stress is applied so that the shear rate increases further, a hydrodynamic coupling between the particles may overcome the interparticle forces responsible for Brownian motion. The particles may be driven closer together, and the microstructure of the colloidal dispersion may change, so that particles cluster together in hydroclusters.

[0050] The viscosity curve of the STF can be fine-tuned through changes in the characteristics of the particles suspended in the solvent. For example, the particles shape, surface chemistry, ionic strength, and size affect the various interparticle forces involved, as does the properties of the solvent. However, in general, hydrodynamic forces dominate at a high shear stress, which also makes the addition of a polymer attached to the particle surface effective in limiting clumping in hydroclusters. Various factors influence this clumping behavior, including, fluid slip, adsorbed ions, surfactants, polymers, surface roughness, graft density (e.g., of a grafted polymer), molecular weight, and solvent, so that the onset of shear thickening can be modified. In general, the onset of shear thickening can be slowed by the introduction of techniques to prevent the clumping of particles. For example, influencing the STF with emissions from an emitter in proximal location to the chamber.

[0051] FIG. **1**C is a graph of piston velocity vs. force applied to the piston for an aspect of an embodiment of a mechanical and computing system that includes a chamber, a shear thickening fluid, and a piston that moves through the chamber applying forces on the shear thickening fluid.

The shear thickening fluid includes a non-Newtonian fluid since the relationship between shear rate and viscosity is nonlinear.

[0052] An example curve for a shear thickening fluid indicates that as more force is applied to the piston in zone A, a higher piston velocity is realized until the corresponding transition to zone B occurs where the shear threshold affect takes hold and the viscosity abruptly increases significantly. When the viscosity increases abruptly, the piston velocity slows back down and may even stop.

[0053] Another example curve for a diluted shear thickening fluid indicates that as more force is applied to the piston in zone A, an even higher piston velocity is realized until the corresponding transition to zone B occurs where the shear threshold affect takes hold and the viscosity abruptly increases significantly. When the viscosity increases abruptly, the piston velocity slows back down and may even stop.

[0054] FIG. **2**A is a schematic block diagram of an embodiment of the computing entity (e.g., **20-1** through **20**-N; and **22**) of the mechanical and computing system of FIG. **1**. The computing entity includes one or more computing devices **100-1** through **100**-N. A computing device is any electronic device that communicates data, processes data, represents data (e.g., user interface) and/or stores data.

[0055] Computing devices include portable computing devices and fixed computing devices. Examples of portable computing devices include an embedded controller, a smart sensor, a social networking device, a gaming device, a smart phone, a laptop computer, a tablet computer, a video game controller, and/or any other portable device that includes a computing core. Examples of fixed computing devices includes a personal computer, a computer server, a cable set-top box, a fixed display device, an appliance, and industrial controller, a video game counsel, a home entertainment controller, a critical infrastructure controller, and/or any type of home, office or cloud computing equipment that includes a computing core.

[0056] FIG. **2**B is a schematic block diagram of an embodiment of a computing device (e.g., **100-1** through **100**-N) of the computing entity of FIG. **2**A that includes one or more computing cores **52-1** through **52**-N, a memory module **102**, a human interface module **18**, an environment sensor module **14**, and an input/output (I/O) module **104**. In alternative embodiments, the human interface module **18**, the environment sensor module **14**, the I/O module **104**, and the memory module **102** may be standalone (e.g., external to the computing device). An embodiment of the computing device is discussed in greater detail with reference to FIG. **3**.

[0057] FIG. **3** is a schematic block diagram of another embodiment of the computing device **100-1** of the mechanical and computing system of FIG. **1** that includes the human interface module **18**, the environment sensor module **14**, the computing core **52-1**, the memory module **102**, and the I/O module **104**. The human interface module **18** includes one or more visual output devices **74** (e.g., video graphics display, 3-D viewer, touchscreen, LED, etc.), one or more visual input devices **80** (e.g., a still image camera, a video camera, a 3-D video camera, photocell, etc.), and one or more audio output devices **78** (e.g., speaker(s), headphone jack, a motor, etc.). The human interface module **18** further includes one or more user input devices **76** (e.g., keypad, keyboard, touchscreen, voice to text, a push button, a microphone, a card reader, a door position switch, a biometric input device, etc.) and one or more motion output devices **106** (e.g., servos, motors, lifts, pumps, actuators, anything to get real-world objects to move).

[0058] The computing core **52-1** includes a video graphics module **54**, one or more processing modules **50-1** through **50**-N, a memory controller **56**, one or more main memories **58-1** through **58**-N (e.g., RAM), one or more input/output (I/O) device interface modules **62**, an input/output (I/O) controller **60**, and a peripheral interface **64**. A processing module is as defined at the end of the detailed description.

[0059] The memory module **102** includes a memory interface module **70** and one or more memory devices, including flash memory devices **92**, hard drive (HD) memory **94**, solid state (SS) memory **96**, and cloud memory **98**. The cloud memory **98** includes an on-line storage system and an on-line backup system.

[0060] The I/O module **104** includes a network interface module **72**, a peripheral device interface module **68**, and a universal serial bus (USB) interface module **66**. Each of the I/O device interface module **62**, the peripheral interface **64**, the memory interface module **70**, the network interface module **72**, the peripheral device interface module **68**, and the USB interface modules **66** includes a combination of hardware (e.g., connectors, wiring, etc.) and operational instructions stored on memory (e.g., driver software) that are executed by one or more of the processing modules **50-1** through **50-**N and/or a processing circuit within the particular module.

[0061] The I/O module **104** further includes one or more wireless location modems **84** (e.g., global positioning satellite (GPS), Wi-Fi, angle of arrival, time difference of arrival, signal strength, dedicated wireless location, etc.) and one or more wireless communication modems **86** (e.g., a cellular network transceiver, a wireless data network transceiver, a Wi-Fi transceiver, a Bluetooth transceiver, a 315 MHz transceiver, a zig bee transceiver, a 60 GHz transceiver, etc.). The I/O module **104** further includes a telco interface **108** (e.g., to interface to a public switched telephone network), a wired local area network (LAN) **88** (e.g., optical, electrical), and a wired wide area network (WAN) **90** (e.g., optical, electrical). The I/O module **104** further includes one or more peripheral devices (e.g., peripheral devices **1**-P) and one or more universal serial bus (USB) devices (USB devices **1**-U). In other embodiments, the computing device **100-1** may include more or less devices and modules than shown in this example embodiment.

[0062] FIG. **4** is a schematic block diagram of an embodiment of the environment sensor module **14** of the computing device of FIG. **2**B that includes a sensor interface module **120** to output environment sensor information **150** based on information communicated with a set of sensors. The set of sensors includes a visual sensor **122** (e.g., to the camera, 3-D camera, 360° view camera, a camera array, an optical spectrometer, etc.) and an audio sensor **124** (e.g., a microphone, a microphone array). The set of sensors further includes a motion sensor **126** (e.g., a solid-state Gyro, a vibration detector, a laser motion detector) and a position sensor **128** (e.g., a Hall effect sensor, an image detector, a GPS receiver, a radar system).

[0063] The set of sensors further includes a scanning sensor **130** (e.g., CAT scan, MRI, x-ray, ultrasound, radio scatter, particle detector, laser measure, further radar) and a temperature sensor **132** (e.g., thermometer, thermal coupler). The set of sensors further includes a humidity sensor **134** (resistance based, capacitance based) and an altitude sensor **136** (e.g., pressure based, GPS-based, laser-based).

[0064] The set of sensors further includes a biosensor **138** (e.g., enzyme, microbial) and a chemical sensor **140** (e.g., mass spectrometer, gas, polymer). The set of sensors further includes a magnetic sensor **142** (e.g., Hall effect, piezo electric, coil, magnetic tunnel junction) and any generic sensor **144** (e.g., including a hybrid combination of two or more of the other sensors).

[0065] FIGS. **5**A-**5**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system includes a head unit system that includes the head unit **10-1** of FIG. **1**, the object **12-1** of FIG. **1**, the environment sensor module **14** of FIG. **2**B, and the computing entity **20-1** of FIG. **1**.

[0066] The head unit system further includes an environment sensor (e.g., environment sensor module **14** of FIG. **2**B). The environment sensor is associated with an external environment that is external (e.g., outdoors) to an internal environment associated with the object. For example, the internal environment includes facilities inside a building and the external environment includes the environment outside the building.

[0067] The head unit **10-1** includes a shear thickening fluid (STF) **42**. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. **1**B. The second range of shear rates are greater than the first range of shear rates.

[0068] The head unit further includes a chamber **16**. The chamber is configured to contain a portion of the STF and includes a front channel **26** and a back channel **24**.
[0069] The head unit further includes a piston **36** housed at least partially radially within the chamber **16** and separating the back channel **24** and the front channel **26**. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object **12-1**. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.
[0070] The piston **36** includes a first piston bypass **38-1** between opposite sides of the piston that controls flow of the STF **42** between the opposite sides of the piston from the back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.
[0071] The piston **36** further includes a second piston bypass **38-2** between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.
[0072] The head unit **10-1** further includes a set of fluid flow sensors **116-1-1** and **116-1-2** positioned proximal to the chamber **16**. The set of fluid flow sensors provide the fluid response **232-1-1** and **232-1-2** respectively from the STF **42**.
[0073] The head unit **10-1** further includes set of fluid manipulation emitters **114-1-1** and **114-1-2** positioned proximal to the chamber **16**. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF **42** (e.g., shifting the shear rate versus viscosity curve), the first piston bypass **38-1** (e.g., to block or allow flow of the STF), and the second piston bypass **38-2** to control the motion of the object **12-1**.
[0074] FIG. **5**A illustrates an example of operation of a method for the controlling the operational aspects that includes the piston **36** moving inward towards the head unit **10-1** when the object **12-1** exerts the force on the plunger **28** that transfers the force to the piston **36**. As a result, the piston **36** exerts the force on the STF **42** within the back channel **24**.
[0075] A first step of the example of operation includes the computing entity **20-1** interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity **182** and a piston position **184** of the piston **36** associated with the head unit device of the head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object **12-1** within the internal environment. For example, the computing entity **20-1** interprets fluid responses **232-1-1** and **232-1-2** from the STF **42** in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.
[0076] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity **20-1** receives fluid responses **232-1-1** and **232-1-2** over the time range, where the fluid responses include the fluid flow signals.
[0077] A second sub-step includes determining the fluid flow response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity **20-1** interprets the fluid flow signals to produce the fluid flow response.
[0078] A third sub-step includes determining the piston velocity based on the fluid flow response of the set of fluid flow sensors over the time range. For example, the computing entity **20-1** calculates piston velocity based on changes in the fluid flow response over the time range.
[0079] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity **20-1** calculates the piston position based on

time in the piston velocity as the piston moves through the chamber.

[0080] As yet another example of interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** compares the fluid response **232-1-1** and **232-1-2** to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity **182** and piston position **184**. As a still further example of the interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** extracts the piston velocity **182** and the piston position **184** directly from the fluid response **232-1-1** and/or **232-1-2** when the sensors **116-1-1** and **116-1-2** generate the piston velocity and piston position directly.

[0081] The first step of the example of operation further includes the computing entity **20-1** determining a shear force **186** based on the piston velocity **182** and the piston position **184**. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid flow response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity **20-1** extracts the shear force **186** directly from the fluid responses **232-1-1** and **232-1-2**. In an instance, the shear force **186** reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. **5**A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y**1**.

[0082] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42**.

[0083] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42** based on an actual valve opening status of the first piston bypass **38-1** (e.g., which allows flow of the STF from the back channel **24** to the front channel **26** when the piston is moving in the inward direction of the example).

[0084] Having determined the piston velocity and the piston position, a second step of the example of operation includes the computing entity **20-1** interpreting an output of the environment sensor to identify an external factor of concern associated with the external environment that is likely to affect the internal environment. The external factor **149** of concern is associated with one or more of a wind gust, wind from a particular direction, rain, snow, humidity, air pressure, temperature, an air pollutant, and/or any other factor of concern. The interpreting the output of the environment sensor to identify the external factor of concern associated with the external environment that is likely to affect the internal environment includes a series of sub-steps.

[0085] A first sub-step includes the computing entity **20-1** obtaining environment sensor information **150** from the environment sensor **14** for a set of timeframes (e.g., every second for 5 minutes). A second sub-step includes the computing entity **20-1** identifying an external factor of external factors **149** from the environment sensor information **150** for the set of timeframes. For example, periodic wind directions and/or periodic external temperatures.

[0086] A third sub-step includes the computing entity **20-1** determining whether an amplitude of the external factor for the set of timeframes is greater than a maximum amplitude threshold level. For example, the computing entity **20-1** interprets environment sensor information **150** to identify external wind direction from the northwest with a wind speed of 75 miles an hour and determining that the wind speed is greater than a maximum wind speed threshold level (e.g., 40 mph).

[0087] A fourth sub-step includes the computing entity **20-1** indicating the external factor of concern associated with the external environment that is likely to affect the internal environment when the amplitude of the external factor for the set of timeframes is greater than the maximum amplitude threshold level. For example, the computing entity **20-1** indicates a low external temperature as external factor of concern when interpreting the environment sensor information **150** to identify an external temperature of 20° below zero Fahrenheit and then determining that the external temperature is less than a minimum external temperature threshold level (e.g., 10° above zero Fahrenheit). The FIG. **5**B further illustrates the example of operation, where having detected the external factor of concern, in a third step the computing entity **20-1** determines the fluid activation **187** for the head unit based on the external factor of concern and one or more of the piston velocity, the piston position, and the shear force **186**. The determining the fluid activation for the head unit device based on the external factor of concern and one or more of the piston velocity and the piston position includes one or more of a variety of approaches.

[0088] A first approach includes the computing entity **20-1** interpreting a request associated with modifying one or more of object velocity and object position. For example, the computing entity **20-1** receives the request from another computing entity with superior information with regards to the external environment and effects on the internal environment.

[0089] A second approach includes the computing entity **20-1** interpreting guidance from a chamber database based on the external factor of concern. For example, the computing entity **20-1** accesses the chamber database **34** FIG. **1**A to recover the guidance for high winds and low temperatures with regards to controlling internal environment objects (e.g., doors).

[0090] A third approach includes the computing entity **20-1** establishing the fluid activation to include facilitating the second range of shear rates to slow down the object when detecting that the external factor of concern includes an external factor that indicates an abatement of the effect on the internal environment from the external factor of concern includes slowing down the object. For example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of an external door more slowly when the external factor indicates an abundance of desirable fresh air.

[0091] A fourth approach includes the computing entity **20-1** establishing the fluid activation to include facilitating the first range of shear rates to speed up the object when detecting that the external factor of concern includes another external factor that indicates the abatement of the effect on the internal environment from the external factor of concern includes speeding up the object. For example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of an external door more quickly, yet safely, when the external factor is the wind speed and direction and the object **12-1** includes the external door. As another example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of the external door more quickly, yet safely, when external factor is low ambient temperature.

[0092] The third step further includes the computing entity **20-1** interpreting the output of the environment sensor to update the external factor of concern associated with the external environment to produce updated external factor of concern associated with the external environment. For example, the computing entity **20-1** reestablishes the external factor of concern 30 minutes later. Having updated the external factor of concern, the computing entity **20-1** updates the fluid activation for the head unit device based on the updated external factor of concern associated with the external environment and one or more of the piston velocity and the piston position. For example, the computing entity **20-1** reestablishes the fluid activation to adapt to the most recent external factor of concern.

[0093] Having determined the fluid activation, a fourth step of the example method of operation includes the computing entity **20-1** activating the set of fluid manipulation emitters in accordance with the fluid activation to control the motion of the object to abate the effect on the internal environment from the external factor of concern. The control of the motion includes one or more of direct manipulation of the STF, facilitation of the first shear threshold effect associated with the first piston bypass, and facilitation of the second shear threshold effect associated with the second piston bypass.

[0094] The activating the set of fluid manipulation emitters in accordance with the fluid activation to control the motion of the object to abate the effect on the internal environment from the external factor of concern includes a variety of approaches. When the piston is traveling through the chamber in the inward direction and when the STF is to have the decreasing viscosity, a first sub-approach includes the computing entity **20-1** issuing the fluid activation **234-1-1** to the set of fluid manipulation emitters **114-1-1** to cause one of the first piston bypass **38-1** to facilitate the first shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF **42** to facilitate the first range of shear rates (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0095] When the piston is traveling through the chamber in the inward direction and when the STF is to have the increasing viscosity, a second sub-approach includes the computing entity **20-1** issuing the fluid activation **234-1-2** to the set of fluid manipulation emitters **114-1-2** to cause one of the first piston bypass **38-1** to facilitate the first shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF **42** to facilitate the second range of shear rates. For example, when the object **12-1** includes the external door closing (e.g., piston moving in the inward direction) in a high wind situation, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-1** to facilitate closing down the one-way check valve to prevent STF from moving from the back channel **24** to the front channel **26** thusly selecting the second range of shear rates and a higher viscosity the STF to slow down the door to close safely moving from the point Y**1** to the point Y**2** (e.g., at a dead stop when closed) as illustrated in FIG. **5**B.

[0096] When the piston is traveling through the chamber in the outward direction and when the STF is to have the decreasing viscosity a third sub-approach includes the computing entity **20-1** issuing the fluid activation **234-1-1** to the set of fluid manipulation emitters **114-1-1** to cause one of the second piston bypass **38-2** to facilitate the second shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF **42** to facilitate the first range of shear rates.

[0097] When the piston is traveling through the chamber in the outward direction and when the STF is to have the increasing viscosity a fourth sub-approach includes the computing entity **20-1** issuing the fluid activation **234-1-2** to the set of fluid manipulation emitters **114-1-2** to cause one of the second piston bypass **38-2** to facilitate the second shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF **42** to facilitate the second range of shear rates.

[0098] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. **1** can alternatively be performed by other modules of the system of FIG. **1** or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system **10**, cause one or more computing devices of the mechanical and computing system of FIG. **1** to perform any or all of the method steps described above.

[0099] FIGS. **6**A-**6**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system includes the head unit **10-1** of FIG. **1**, the object **12-1** of FIG. **1**, and the computing entity **20-1** of FIG. **1**.

[0100] The head unit system includes an environment sensor (e.g., environment sensor module **14** of FIG. **2**B). The environment sensor is associated with an internal environment that is internal to an external environment associated with the object. For example, the internal environment includes facilities inside a building and the external environment includes the environment outside the building.

[0101] The head unit **10-1** includes a shear thickening fluid (STF) **42**. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. **1**B. The second range of shear rates are greater than the first range of shear rates.

[0102] The head unit further includes a chamber **16**. The chamber is configured to contain a portion of the STF and includes a front channel **26** and a back channel **24**.

[0103] The head unit further includes a piston **36** housed at least partially radially within the chamber **16** and separating the back channel **24** and the front channel **26**. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object **12-1**. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.

[0104] The piston **36** includes a first piston bypass **38-1** between opposite sides of the piston that controls flow of the STF **42** between the opposite sides of the piston from the back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.

[0105] The piston **36** further includes a second piston bypass **38-2** between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.

[0106] The head unit **10-1** further includes a set of fluid flow sensors **116-1-1** and **116-1-2** positioned proximal to the chamber **16**. The set of fluid flow sensors provide the fluid response **232-1-1** and **232-1-2** respectively from the STF **42**.

[0107] The head unit **10-1** further includes set of fluid manipulation emitters **114-1-1** and **114-1-2** positioned proximal to the chamber **16**. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF **42** (e.g., shifting the shear rate versus viscosity curve), the first piston bypass **38-1** (e.g., to block or allow flow of the STF), and the second piston bypass **38-2** to control the motion of the object **12-1**.

[0108] FIG. **6**A illustrates an example of operation of a method for the controlling the operational aspects. A first step of the example of operation includes the piston **36** moving inward towards the head unit **10-1** when the object **12-1** exerts the force on the plunger **28** that transfers the force to the piston **36**. As a result, the piston **36** exerts the force on the STF **42** within the back channel **24**.

[0109] A second step of the example of operation includes the computing entity **20-1** interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity **182** and a piston position **184** of the piston **36** associated with a head unit device of a head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object **12-1** within the internal environment. For example, the computing entity **20-1** interprets fluid responses **232-1-1** and **232-1-2** from the STF **42** in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0110] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity **20-1** receives fluid responses **232-1-1** and **232-1-2** over the time range, where the fluid responses include the fluid flow signals.

[0111] A second sub-step includes determining the fluid flow response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity **20-1** interprets the fluid flow signals to produce the fluid flow response.

[0112] A third sub-step includes determining the piston velocity based on the fluid flow response of the set of fluid flow sensors over the time range. For example, the computing entity **20-1** calculates piston velocity based on changes in the fluid flow response over the time range.

[0113] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity **20-1** calculates the piston position based on time in the piston velocity as the piston moves through the chamber.

[0114] As yet another example of interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** compares the fluid response **232-1-1** and **232-1-2** to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity **182** and piston position **184**. As a still further example of the interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** extracts the piston velocity **182** and the piston position **184** directly from the fluid response **232-1-1** and/or **232-1-2** when the sensors **116-1-1** and **116-1-2** generate the piston velocity and piston position directly.

[0115] The second step of the example of operation further includes the computing entity **20-1** determining a shear force **186** based on the piston velocity **182** and the piston position **184**. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid flow response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity **20-1** extracts the shear force **186** directly from the fluid responses **232-1-1** and **232-1-2**. In an instance, the shear force **186** reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. **6**A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y**1**.

[0116] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42**.

[0117] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42** based on an actual valve opening status of the first piston bypass **38-1** (e.g., which allows flow of the STF from the back channel **24** to the front channel **26** when the piston is moving in the inward direction of the example).

[0118] A third step of the example of operation includes the computing entity **20-1** interpreting an output of the environment sensor to identify an internal factor **151** of concern associated with the internal environment. The internal factor **151** of concern is associated with one or more of internal air quality, an internal smoke level, date, time of day, a building activity schedule, internal humidity, internal air pressure, internal temperature, fire alarm status, intrusion alarm status, building lockdown status, a building occupancy level, a maximum building occupancy threshold level, building power status, a solar array status, a heating-ventilation-and-cooling (HVAC) status, and/or any other factor of concern. For example, the computing entity **20-1** interprets environment sensor information **150** to identify smoke in a storage room. As another example, the computing entity **20-1** interprets the environment sensor information **150** to identify a scheduled lunch time building activity (e.g., where more than average number of people are moving about within the building).

[0119] FIG. **6**B further illustrates the example of operation, where having detected the internal factor of concern, in a fourth step the computing entity **20-1** determines the fluid activation **187** for the head unit based on the internal factor of concern and one or more of the piston velocity, the piston position, and the shear force **186**. For example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of an internal door associated with the storage room more quickly when the internal factor is the smoke detected in the storage room and the object **12-1** includes the internal door to the storage room. As another example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of the external door slower than average, and safely, when the scheduled lunch time building activity is detected as the internal factor of concern.

[0120] In particular, the computing entity **20-1** determines the fluid activation **187** to adjust the viscosity of the STF to facilitate movement of the piston and hence door in a more desirable fashion based on the identified internal factor of concern. The determining the fluid activation **187** includes a variety of approaches. A first approach includes opening of either of the piston bypass **38-1** and piston bypass **38-2** allow the STF to move between the back channel **24** and the front channel **26** to lower the shear rate and thus select a lower viscosity which in turn allows more rapid movement of the piston in the chamber and hence speeds up the door. A second approach includes opening of the chamber bypass **40** to lower the viscosity of the STF. A third approach includes activating the set of emitters to directly alter the viscosity of the STF in a desired fashion (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0121] A fifth step of the example method of operation includes the computing entity **20-1** activating the set of fluid manipulation emitters **114-1-1** and **114-1-2** in accordance with the fluid activation **187** to manipulate one of the first shear threshold effect associated with the first piston bypass **38-1** and the second shear threshold effect associated with the second piston bypass **38-2** to control the motion of the object **12-1** to abate the effect on the internal environment from the internal factor of concern. For example, when the object **12-1** includes the internal door to the storage room and is closing when moving in the inward direction, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-1** to facilitate further opening of a one-way check valve to allow more of the STF to move from the back channel **24** to the front channel **26** thusly selecting the first range of shear rates and a lower viscosity of the STF to speed up the door to close when the internal factor of concern is the detected smoke in the storage room.

[0122] As another example, when the object **12-1** includes the external door closing (e.g., piston moving in the inward direction) in a high-traffic lunchtime situation, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-1** to facilitate closing down the one-way check valve to prevent STF from moving from the back channel **24** to the front channel **26** thusly selecting the second range of shear rates and a higher viscosity the STF to slow down the door to close safely moving from the point Y**1** to the point Y**2** (e.g., at a dead stop when closed and no one is coming through the door) as illustrated in FIG. **6**B.

[0123] In yet another embodiment, the computing entity **20-1** detects both an external factor of concern and the internal factor of concern to produce the fluid activation **187**. For example, even when the external temperature is very low, the computing entity **20-1** controls the head unit system to close the door slowly when a higher than normal traffic pattern of people going through that door is detected.

[0124] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. **1** can alternatively be performed by other modules of the system of FIG. **1** or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system **10**, cause one or more computing devices of the mechanical and computing system of FIG. **1** to perform any or all of the method steps described above.

[0125] FIGS. **7**A-**7**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system includes the head unit **10-1** of FIG. **1**, the object **12-1** of FIG. **1** (e.g., a door), a secondary object **12-2** (e.g., a person), and the computing entity **20-1** of FIG. **1**.
[0126] The head unit system includes a secondary object sensor (e.g., environment sensor module **14** of FIG. **2**B). The secondary object is associated with the object (e.g., the person goes through the door).
[0127] The head unit **10-1** includes a shear thickening fluid (STF) **42**. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. **1**B. The second range of shear rates are greater than the first range of shear rates.
[0128] The head unit further includes a chamber **16**. The chamber is configured to contain a portion of the STF and includes a front channel **26** and a back channel **24**.
[0129] The head unit further includes a piston **36** housed at least partially radially within the chamber **16** and separating the back channel **24** and the front channel **26**. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object **12-1**. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.
[0130] The piston **36** includes a first piston bypass **38-1** between opposite sides of the piston that controls flow of the STF **42** between the opposite sides of the piston from the back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.
[0131] The piston **36** further includes a second piston bypass **38-2** between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.
[0132] The head unit **10-1** further includes a set of fluid flow sensors **116-1-1** and **116-1-2** positioned proximal to the chamber **16**. The set of fluid flow sensors provide the fluid response **232-1-1** and **232-1-2** respectively from the STF **42**.
[0133] The head unit **10-1** further includes set of fluid manipulation emitters **114-1-1** and **114-1-2** positioned proximal to the chamber **16**. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF **42** (e.g., shifting the shear rate versus viscosity curve), the first piston bypass **38-1** (e.g., to block or allow flow of the STF), and the second piston bypass **38-2** to control the motion of the object **12-1** with regards to the secondary object (e.g., avoiding a collision between the door and the person).
[0134] FIG. **7**A illustrates an example of operation of a method for the controlling the operational aspects. A first step of the example of operation includes the piston **36** moving outward away from the head unit **10-1** when the object **12-1** exerts a pulling force on the plunger **28** that transfers the force to the piston **36**. As a result, the piston **36** exerts the force on the STF **42** within the front channel **26**.
[0135] A second step of the example of operation includes the computing entity **20-1** interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity **182** and a piston position **184** of the piston **36** associated with a head unit device of a head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object **12-1** within the internal environment. For example, the computing entity **20-1** interprets fluid responses **232-1-1** and **232-1-2** from the STF **42** in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0136] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity **20-1** receives fluid responses **232-1-1** and **232-1-2** over the time range, where the fluid responses include the fluid flow signals.
[0137] A second sub-step includes determining the fluid flow response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity **20-1** interprets the fluid flow signals to produce the fluid response.
[0138] A third sub-step includes determining the piston velocity based on the fluid response of the set of fluid flow sensors over the time range. For example, the computing entity **20-1** calculates piston velocity based on changes in the fluid response over the time range.
[0139] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity **20-1** calculates the piston position based on time in the piston velocity as the piston moves through the chamber.
[0140] As yet another example of interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** compares the fluid response **232-1-1** and **232-1-2** to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity **182** and piston position **184**. As a still further example of the interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** extracts the piston velocity **182** and the piston position **184** directly from the fluid response **232-1-1** and/or **232-1-2** when the sensors **116-1-1** and **116-1-2** generate the piston velocity and piston position directly.
[0141] The second step of the example of operation further includes the computing entity **20-1** determining a shear force **186** based on the piston velocity **182** and the piston position **184**. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity **20-1** extracts the shear force **186** directly from the fluid responses **232-1-1** and **232-1-2**. In an instance, the shear force **186** reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. **7**A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y**3** (e.g., a negative velocity since moving in the outward direction).
[0142] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42**.
[0143] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42** based on an actual valve opening status of the first piston bypass **38-1** (e.g., which allows flow of the STF from the front channel **26** to the back channel **24** when the piston is moving in the outward direction of the example).
[0144] A third step of the example of operation includes the computing entity **20-1** interpreting an output of the secondary object sensor to produce an object type of the secondary object. The object type includes a person, a child, an elderly person, a group of people, a patient transport gurney, a cart, a short cart, a long cart, a cart train, a motor scooter, a vehicle, a jack truck, a pallet hauler, a 2 wheel hauler, a set of animals, and/or type of anything that can move through a door. The interpreting includes interpreting environment sensor information **150** from the secondary object sensor (e.g., from the environment sensor module **14**), comparing the environment sensor information **150** to previously stored information for each of the object types, and selecting the

object type when a match is detected. For example, the computing entity **20-1** compares an image of the environment sensor information **152** a stored image of a cart and identifies the secondary object as the cart when matching the image of the environment sensor information **150** to the stored image.

[0145] FIG. **7**B further illustrates the example of operation, where having determined the object type, in a fourth step the computing entity **20-1** determines the fluid activation **187** for the head unit based on the object type and one or more of the piston velocity, the piston position, and the shear force **186**. For example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of an external door more quickly, yet safely, when the object type is a short cart. As another example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of the external door more slowly when the object type is an elderly person.

[0146] In particular, the computing entity **20-1** determines the fluid activation **187** to adjust the viscosity of the STF to facilitate movement of the piston and hence door in a more desirable fashion based on the identified object type. The determining the fluid activation **187** includes a variety of approaches. A first approach includes opening of either of the piston bypass **38-1** and piston bypass **38-2** allow the STF to move between the back channel **24** and the front channel **26** to lower the shear rate and thus select a lower viscosity which in turn allows more rapid movement of the piston in the chamber and hence speeds up the door. A second approach includes opening of the chamber bypass **40** to lower the viscosity of the STF. A third approach includes activating the set of emitters to directly alter the viscosity of the STF in a desired fashion (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0147] A fifth step of the example method of operation includes the computing entity **20-1** activating the set of fluid manipulation emitters **114-1-1** and **114-1-2** in accordance with the fluid activation **187** to manipulate one of the first shear threshold effect associated with the first piston bypass **38-1** and the second shear threshold effect associated with the second piston bypass **38-2** to control the motion of the object **12-1** to control the motion of the object with regards to the secondary object. For example, when the object **12-1** includes an operating room door and is opening when moving in the outward direction, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-2** to facilitate further opening of a one-way check valve to allow more of the STF to move from the front channel **26** to the back channel **24** thusly selecting the first range of shear rates and a lower viscosity of the STF to speed up the door to open when the detected secondary object is a patient transport gurney entering the operating room.

[0148] As another example, when the object **12-2** includes the group of people, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-1** to facilitate closing down the one-way check valve to prevent STF from moving from the back channel **24** to the front channel **26** thusly selecting the second range of shear rates and a higher viscosity the STF to slow down the door to close safely moving from the point Y**3** to the point Y**4** (e.g., when the piston is moving in the outward direction) as illustrated in FIG. **7**B.

[0149] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. **1** can alternatively be performed by other modules of the system of FIG. **1** or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system **10**, cause one or more computing devices of the mechanical and computing system of FIG. **1** to perform any or all of the method steps described above.

[0150] FIGS. **8**A-**8**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system provides a head unit system that includes the head unit **10-1** of FIG. **1**, the object **12-1** of FIG. **1** (e.g., a door), a plurality of secondary objects **12-2** through **12**-N, a secondary object sensor (e.g., the environment sensor module **14** of FIG. **2**B to detect the secondary objects), and the computing entity **20-1** of FIG. **1**. The secondary object sensor is associated with the object **12-1** and the secondary objects are associated with the object.

[0151] The head unit system includes the secondary object sensor. The plurality of secondary objects is associated with the object (e.g., a group of people go through the door).

[0152] The head unit **10-1** includes a shear thickening fluid (STF) **42**. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. **1**B. The second range of shear rates are greater than the first range of shear rates.

[0153] The head unit further includes a chamber **16**. The chamber is configured to contain a portion of the STF and includes a front channel **26** and a back channel **24**.

[0154] The head unit further includes a piston **36** housed at least partially radially within the chamber **16** and separating the back channel **24** and the front channel **26**. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object **12-1**. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.

[0155] The piston **36** includes a first piston bypass **38-1** between opposite sides of the piston that controls flow of the STF **42** between the opposite sides of the piston from the back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.

[0156] The piston **36** further includes a second piston bypass **38-2** between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.

[0157] The head unit **10-1** further includes a set of fluid flow sensors **116-1-1** and **116-1-2** positioned proximal to the chamber **16**. The set of fluid flow sensors provide the fluid response **232-1-1** and **232-1-2** respectively from the STF **42**.

[0158] The head unit **10-1** further includes set of fluid manipulation emitters **114-1-1** and **114-1-2** positioned proximal to the chamber **16**. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF **42** (e.g., shifting the shear rate versus viscosity curve), the first piston bypass **38-1** (e.g., to block or allow flow of the STF), and the second piston bypass **38-2** to control the motion of the object **12-1** with regards to the set of secondary objects (e.g., avoiding a collision between the door and the group people).

[0159] FIG. **8**A illustrates an example of operation of a method for the controlling the operational aspects. In the example of operation, the piston **36** moves outward away from the head unit **10-1** when the object **12-1** exerts a pulling force on the plunger **28** that transfers the force to the piston **36**. As a result, the piston **36** exerts the force on the STF **42** within the front channel **26**.

[0160] A first step of the example of operation includes the computing entity **20-1** interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity **182** and a piston position **184** of the piston **36** associated with a head unit device of a head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object **12-1**. For example, the computing entity **20-1** interprets fluid responses **232-1-1** and **232-1-2** from the STF **42** in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0161] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity **20-1** receives fluid responses **232-1-1** and **232-1-2** over the time range, where the fluid responses include the fluid flow signals.
[0162] A second sub-step includes determining the fluid flow response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity **20-1** interprets the fluid flow signals to produce the fluid response.
[0163] A third sub-step includes determining the piston velocity based on the fluid response of the set of fluid flow sensors over the time range. For example, the computing entity **20-1** calculates piston velocity based on changes in the fluid response over the time range.
[0164] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity **20-1** calculates the piston position based on time in the piston velocity as the piston moves through the chamber.
[0165] As yet another example of interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** compares the fluid response **232-1-1** and **232-1-2** to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity **182** and piston position **184**. As a still further example of the interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** extracts the piston velocity **182** and the piston position **184** directly from the fluid response **232-1-1** and/or **232-1-2** when the sensors **116-1-1** and **116-1-2** generate the piston velocity and piston position directly.
[0166] The first step of the example of operation further includes the computing entity **20-1** determining a shear force **186** based on the piston velocity **182** and the piston position **184**. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity **20-1** extracts the shear force **186** directly from the fluid responses **232-1-1** and **232-1-2**. In an instance, the shear force **186** reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. **8**A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y**3** (e.g., a negative velocity since moving in the outward direction).
[0167] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42**.
[0168] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42** based on an actual valve opening status of the second piston bypass **38-2** (e.g., which allows flow of the STF from the front channel **26** to the back channel **24** when the piston is moving in the outward direction of the example).
[0169] A second step of the example of operation includes the computing entity **20-1** interpreting an output of the secondary object sensor to update an object pattern of the plurality of secondary objects. The object pattern includes the object type as previously discussed and activity parameters associated with objects of the object type. The activity parameters includes a person walking, a person running, a group of people walking, a group of people running, a set of secondary objects moving towards the object, the set of second area objects moving away from the object, and/or any other possible activity type associated with one or more of the secondary objects.
[0170] The interpreting the output of the secondary object sensor to update the object pattern of the

plurality of secondary objects to produce the updated object pattern of the plurality of secondary objects includes a series of sub-steps. A first sub-step includes the computing entity **20-1** obtaining environment sensor information **150** from the secondary object sensor (e.g., environment sensor module **14**) for a subset of secondary objects of the plurality of secondary objects (e.g., **12-2** through **12**-N).

[0171] A second sub-step includes the computing entity **20-1** identifying activity parameters from the environment sensor information for the subset of secondary objects. For example, the computing entity **20-1** compares a video clip of the environment sensor information **150** to a stored video clip of a group of people walking towards a door and identifies the object pattern as a group of people walking when matching the video clip of the environment sensor information **150** to the stored video clip of the person walking. As another example, the computing entity **20-1** further determines walking velocities for the group of people from the video clip from the video clip.

[0172] A third sub-step includes the computing entity **20-1** obtaining the object pattern of the plurality of secondary objects. For example, the computing entity **20-1** recovers the object pattern from a local memory of the computing entity **20-1**. As another example, the computing entity **20-1** extracts the object pattern from activity information **153** recovered from the chamber database **34**.

[0173] A fourth sub-step includes the computing entity **20-1** modifying the object pattern of the plurality of secondary objects based on the activity parameters for the subset of secondary objects to produce the updated object pattern of the plurality of secondary objects. For example, the computing entity **20-1** modifies the walking velocities for the group of people based on the recently extracted walking velocities for the group of people.

[0174] FIG. **8**B further illustrates the example of operation, where having determined the updated object pattern for the plurality of secondary objects, a third step includes the computing entity **20-1** determining fluid activation **187** for the head unit based on the updated object pattern of the plurality of secondary objects and one or more of activity information **153**, the piston velocity **182**, the piston position **184**, and the shear force **186**. The activity information **153** includes historical records and schedules for future activities associated with the set of secondary objects. For example, the activity information **153** includes a work schedule, a class schedule, timeclock punching information indicating which employees are present at an area of employment, invoicing information associated with product flow, manufacturing process information, inventory control information, and/or any other information that assists in identifying the object pattern for the set of secondary objects.

[0175] The determining the fluid activation for the head unit device is based on the updated object pattern of the plurality of secondary objects and one or more of the piston velocity and the piston position includes one or more sub-steps. A first sub-step includes the computing entity **20-1** interpreting a request associated with modifying one or more of object velocity and object position. For example, the computing entity **20-1** receives the request from another computing entity. A second sub-step includes the computing entity interpreting fluid activation guidance from the chamber database **34** based on the updated object pattern of the plurality of secondary objects. For example, the computing entity **20-1** accesses the chamber database **34** based on the updated object pattern to locate the fluid activation guidance.

[0176] A third sub step includes the computing entity **20-1** interpreting activity information **153** from the chamber database **34** based on the updated object pattern of the plurality of secondary objects to produce an object movement recommendation. A correlation between the activity information and the updated object pattern of the plurality of secondary objects suggests an expected movement behavior of the secondary object. For example, analysis of **100** plus instances of groups of people walking through doors is summarized in the chamber database as the activity information **153** such that one more instance of the group of people walking through a particular door is expected to be similar.

[0177] A fourth sub-step includes the computing entity **20-1** determining a position for the secondary object based on the updated object pattern of the plurality of secondary objects. For example, the computing entity **20-1** estimates a next position for a person walking through the door based on the pattern of the groups of people walking through the doors.

[0178] A fifth sub-step includes the computing entity **20-1** determining an object position for the object based on the piston velocity and the piston position. For example, the computing entity **20-1** estimates the object position for the object (e.g., a door with regards to the secondary object, a person) based on historical data of piston velocity and piston position verses position of the object (e.g., the door).

[0179] A sixth sub-step includes the computing entity **20-1** establishing the fluid activation to include facilitating the first range of shear rates when the object movement recommendation includes increasing velocity of the object. For example, when the piston is traveling in the outward direction to open the door, the first range of shear rates is selected when the object movement recommendation includes opening the door more quickly to avoid a collision between the person and the door.

[0180] A seventh sub-step includes the computing entity **20-1** establishing the fluid activation to include facilitating the second range of shear rates when the object movement recommendation includes decreasing the velocity of the object. For example, when the piston is traveling in the outward direction to close the door, the second range of shear rates is selected when the object movement recommendation includes closing the door more slowly to avoid the collision between the person and the door. As another example of the determining of the fluid activation **187**, the computing entity **20-1** establishes the fluid activation **187** to facilitate opening of a large door more quickly, yet safely, when the object pattern is the group of people approaching the door and the activity information **153** indicates that a current time frame is a lunch. When large groups of people are expected to be moving through the door. As a still further example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of the large door more slowly when the object pattern is the group of people passing through the door.

[0181] In particular, the computing entity **20-1** determines the fluid activation **187** to produce a mechanism to adjust the viscosity of the STF to facilitate movement of the piston and hence door in a more desirable fashion based on the identified object type. The mechanism to adjust the viscosity includes a variety of approaches. A first approach includes opening of either of the piston bypass **38-1** and piston bypass **38-2** allow the STF to move between the back channel **24** and the front channel **26** to lower the shear rate and thus select a lower viscosity which in turn allows more rapid movement of the piston in the chamber and hence speeds up the door. A second approach includes opening of the chamber bypass **40** to lower the viscosity of the STF. A third approach includes activating the set of emitters to directly alter the viscosity of the STF in a desired fashion (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0182] The determining the fluid activation further includes interpreting the output of the secondary object sensor to further update the object pattern of the plurality of secondary objects to produce a further updated object pattern of the plurality of secondary objects. For example, the computing entity **20-1** further analyzes environment sensor information **150** from the environment sensor module **14** to update the object pattern. Having produced the further updated object pattern, the computing entity **20-1** updates the fluid activation for the head unit device based on the further updated object pattern of the plurality of secondary objects and one or more of the piston velocity and the piston position as previously discussed.

[0183] A fourth step of the example method of operation includes the computing entity **20-1** activating the set of fluid manipulation emitters **114-1-1** and **114-1-2** in accordance with the fluid activation **187** to manipulate one of the first shear threshold effect associated with the first piston bypass **38-1** and the second shear threshold effect associated with the second piston bypass **38-2** to control the motion of the object **12-1** to control the motion of the object with regards to the secondary object. For example, when the object **12-1** includes a large lunchroom door and is opening when moving in the outward direction, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-2** to facilitate further opening of a one-way check valve to allow more of the STF to move from the front channel **26** to the back channel **24** thusly selecting the first range of shear rates and a lower viscosity of the STF to speed up the door to open when the detected set of secondary objects is a group of people passing through the lunch room door during the lunch period.

[0184] As another example, when the set of secondary objects **12-2** through **12**-N includes the group of people, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-2** to facilitate closing down the one-way check valve to prevent STF from moving from the front channel **26** to the back channel **24** thusly selecting the second range of shear rates and a higher viscosity the STF to slow down the door to close safely moving from the point Y**3** to the point Y**4** (e.g., when the piston is moving in the outward direction) as illustrated in FIG. **8**B.

[0185] The activating the set of fluid manipulation emitters in accordance with the fluid activation to control the motion of the object includes a variety of approaches. When the piston is traveling through the chamber in the inward direction and when the STF is to have the decreasing viscosity, a first sub-approach includes the computing entity **20-1** issuing the fluid activation **234-1-1** to the set of fluid manipulation emitters **114-1-1** to cause one of the first piston bypass **38-1** to facilitate the first shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF **42** to facilitate the first range of shear rates (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0186] When the piston is traveling through the chamber in the inward direction and when the STF is to have the increasing viscosity, a second sub-approach includes the computing entity **20-1** issuing the fluid activation **234-1-2** to the set of fluid manipulation emitters **114-1-2** to cause one of the first piston bypass **38-1** to facilitate the first shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF **42** to facilitate the second range of shear rates.

[0187] When the piston is traveling through the chamber in the outward direction and when the STF is to have the decreasing viscosity a third sub-approach includes the computing entity **20-1** issuing the fluid activation **234-1-1** to the set of fluid manipulation emitters **114-1-1** to cause one of the second piston bypass **38-2** to facilitate the second shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF **42** to facilitate the first range of shear rates.

[0188] When the piston is traveling through the chamber in the outward direction and when the STF is to have the increasing viscosity a fourth sub-approach includes the computing entity **20-1** issuing the fluid activation **234-1-2** to the set of fluid manipulation emitters **114-1-2** to cause one of the second piston bypass **38-2** to facilitate the second shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF **42** to facilitate the second range of shear rates.

[0189] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. **1** can alternatively be performed by other modules of the system of FIG. **1** or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system **10**, cause one or more computing devices of the mechanical and computing system of FIG. **1** to perform any or all of the method steps described above.

[0190] FIGS. **9**A-**9**B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system provides a head unit system that includes the head unit **10-1** of FIG. **1**, the object **12-1** of FIG. **1** (e.g., a first door, interchangeable referred to as a first object), a set of objects **12-2** through **12**-N (e.g., people proximal to the first door of a first building passing through the first door), a secondary object sensor (e.g., the environment sensor module **14** of FIG. **2**B to detect the people), the chamber database **34** FIG. **1**, and the computing entities **20-1** through **20**-N (e.g., associated with multiple other non-proximal areas of a large building, associated with a multitude of non-proximal buildings of a large geographic area such as worldwide).

[0191] The head unit **10-1** (e.g., interchangeably referred to as the head unit device) is configured to control motion of the first object (e.g., object **12-1**) with regards to a second object of a plurality of proximal objects (e.g., the objects **12-2** through **12**-N). The plurality of proximal objects are line-of-sight spatially associated with the first object from time to time. For example, a group of people or other objects that are in an immediate vicinity of the first door.

[0192] The head unit system includes the secondary object sensor for sensing the plurality of proximal objects. The head unit **10-1** includes a shear thickening fluid (STF) **42**. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. **1**B. The second range of shear rates are greater than the first range of shear rates.

[0193] The head unit further includes a chamber **16**. The chamber is configured to contain a portion of the STF and includes a front channel **26** and a back channel **24**.

[0194] The head unit further includes a piston **36** housed at least partially radially within the chamber **16** and separating the back channel **24** and the front channel **26**. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object **12-1**. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.

[0195] The piston **36** includes a first piston bypass **38-1** between opposite sides of the piston that controls flow of the STF **42** between the opposite sides of the piston from the back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.

[0196] The piston **36** further includes a second piston bypass **38-2** between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.

[0197] The head unit **10-1** further includes a set of fluid flow sensors **116-1-1** and **116-1-2** positioned proximal to the chamber **16**. The set of fluid flow sensors provide the fluid response **232-1-1** and **232-1-2** respectively from the STF **42**.

[0198] The head unit **10-1** further includes set of fluid manipulation emitters **114-1-1** and **114-1-2** positioned proximal to the chamber **16**. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF **42** (e.g., shifting the shear rate versus viscosity curve), the first piston bypass **38-1** (e.g., to block or allow flow of the STF), and the second piston bypass **38-2** to control the motion of the object **12-1** with regards to the set of secondary objects (e.g., avoiding a collision between the door and the group people).

[0199] FIG. **9**A illustrates an example of operation of a method for the controlling the operational aspects. A first step of the example of operation includes the piston **36** moving outward away from the head unit **10-1** when the object **12-1** exerts a pulling force on the plunger **28** that transfers the force to the piston **36**. As a result, the piston **36** exerts the force on the STF **42** within the front channel **26**.

[0200] A second step of the example of operation includes the computing entity **20-1** interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity **182** and a piston position **184** of the piston **36** associated with the head unit device of the head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object **12-1** with regards to the second object of the plurality of proximal objects (e.g., any one of objects **12-2** through **12**-N). For example, the computing entity **20-1** interprets fluid responses **232-1-1** and **232-1-2** from the STF **42** in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0201] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity **20-1** receives fluid responses **232-1-1** and **232-1-2** over the time range, where the fluid responses include the fluid flow signals.

[0202] A second sub-step includes determining the fluid response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity **20-1** interprets the fluid flow signals to produce the fluid response.

[0203] A third sub-step includes determining the piston velocity based on the fluid response of the set of fluid flow sensors over the time range. For example, the computing entity **20-1** calculates piston velocity based on changes in the fluid response over the time range.

[0204] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity **20-1** calculates the piston position based on time in the piston velocity as the piston moves through the chamber.

[0205] As yet another example of interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** compares the fluid response **232-1-1** and **232-1-2** to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity **182** and piston position **184**. As a still further example of the interpreting the fluid response **232-1-1** and **232-1-2**, the computing entity **20-1** extracts the piston velocity **182** and the piston position **184** directly from the fluid response **232-1-1** and/or **232-1-2** when the sensors **116-1-1** and **116-1-2** generate the piston velocity and piston position directly.

[0206] The second step of the example of operation further includes the computing entity **20-1** determining a shear force **186** based on the piston velocity **182** and the piston position **184**. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity **20-1** extracts the shear force **186** directly from the fluid responses **232-1-1** and **232-1-2**. In an instance, the shear force **186** reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. **9**A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y**3** (e.g., a negative velocity since moving in the outward direction).

[0207] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42**.

[0208] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42** based on an actual valve opening status of the second piston bypass **38-2** (e.g., which allows flow of the STF from the front channel **26** to the back channel **24** when the piston is moving in the outward direction of the example).

[0209] A third step of the example of operation includes the computing entity **20-1** interpreting an output of the secondary object sensor to produce an updated object pattern of the plurality of proximal objects. The object pattern includes the object type as previously discussed and activity parameters associated with objects of the object type. The activity parameters include a person walking, a person running, a group of people walking, a group of people running, a set of secondary objects moving towards the object, the set of second area objects moving away from the object, and/or any other possible activity type associated with one or more of the secondary objects.
[0210] The interpreting the output of the secondary object sensor to produce the updated object pattern of the plurality of proximal objects includes a series of sub-steps. A first sub-step includes obtaining environment sensor information from the secondary object sensor for a subset of the plurality of proximal objects.
[0211] For example, the computing entity **20-1** receives environment sensor information **150** from the environment sensor module **14** with regards to the object **12-2**.
[0212] A second sub-step includes identifying activity parameters from the environment sensor information for the subset of the plurality of proximal objects. For example, the computing entity **20-1** identifies the person walking when the object **12-2** is the person.
[0213] A third sub-step includes obtaining a previous object pattern of the plurality of the plurality of proximal objects. For example, the computing entity **20-1** recovers the previous object pattern for the person walking and for other people walking from the chamber database **34**.
[0214] A fourth sub-step includes modifying the previous object pattern of the plurality of proximal objects based on the activity parameters for the subset of proximal objects to produce the updated object pattern of the plurality of proximal objects. For example, the computing entity **20-1** modifies the previous object pattern to include particulars associated with the person walking.
[0215] FIG. **9**B further illustrates the example the method of operation, where having produced the updated object pattern of the plurality of proximal objects, a fourth step includes the computing entity **20-1** identifying historical motion control results for a multitude of non-proximal objects with regards to a set of comparative objects (e.g., other doors) based on the updated object pattern of the plurality of proximal objects. The multitude of non-proximal objects are non-line-of-sight spatially associated with the set of comparative objects (e.g., people in other buildings across the world). The set of comparative objects share a set of common attributes with the first object (e.g., other doors in the other buildings).
[0216] For example, the computing entity **20-1** compares the updated object pattern to historical results **155** from the chamber database **34**. The other computing entities **20-2** through **20-1** contribute to the historical results by generating the historical results based on other object patterns associated with the other computing entities, corresponding utilized fluid activations for those other object patterns, and actual results from the utilization of the fluid activations (e.g., unfavorable results including late door openings and early door closings; a verbal results including door openings and closings matched with actual needs).
[0217] Having identified the historical motion control results, a fifth step of the example method of operation includes the computing entity **20-1** determining the fluid activation **187** for the head unit based on the historical motion control results for the multitude of non-proximal objects and one or more of the piston velocity, the piston position, and the shear force **186**. The determining the fluid activation for the head unit device based on the historical motion control results for the multitude of non-proximal objects and one or more of the piston velocity and the piston position includes a variety of one or more approaches.
[0218] A first approach includes interpreting a request associated with modifying one or more of object velocity and object position. For example, the computing entity **20-1** receives a request to slow down the door from the computing entity **20-2** when the computing entity **20-2** is associated with a similar door and has produced the request based on a desired optimization.
[0219] A second approach includes interpreting fluid activation guidance from the chamber

database based on the historical motion control results for the multitude of non-proximal objects. For example, the computing entity **20-1** analyzes historical results **155** for similar patterns that lead to desired results of controlling a door similar to the first door.

[0220] A third approach includes interpreting activity information from the chamber database based on the updated object pattern of the plurality of proximal objects to produce an object movement recommendation. A correlation between the activity information and the updated object pattern of the plurality of proximal objects suggests an expected movement behavior of the second object. For example, the computing entity **20-1** produces the object movement recommendation to quickly move the door out of the way when sudden but movement behavior of the object **12-2** is expected based on the activity information.

[0221] A fourth approach includes determining a position for the second object based on the updated object pattern of the plurality of proximal objects. For example, the computing entity **20-1** includes the position for the second object within the updated object pattern for most of the proximal objects.

[0222] A fifth approach includes determining an object position for the first object based on the piston velocity and the piston position. The computing entity **20-1** continuously tracks the object position for the first object based on its velocity and position over time.

[0223] A sixth approach includes establishing the fluid activation to include facilitating the first range of shear rates when the object movement recommendation includes increasing velocity of the first object. For example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of a large exterior door more quickly, yet safely, when the object pattern of the set of secondary objects is one person approaching the door and the historical results **155** indicates that a faster closing for one person is generally favorable.

[0224] A seventh approach includes establishing the fluid activation to include facilitating the second range of shear rates when the object movement recommendation includes decreasing the velocity of the first object. For example, the computing entity **20-1** establishes the fluid activation **187** to facilitate opening of the large exterior door more slowly when the object pattern of the set of secondary objects is a group of people mulling around that could be in the way of the door opening and the historical results **155** indicates that a slower opening when people may be in the way of the opening door is generally favorable.

[0225] In particular, the computing entity **20-1** determines the fluid activation **187** to adjust the viscosity of the STF to facilitate movement of the piston and hence the first door in a more desirable fashion based on the identified object pattern for the set of proximal objects and based on the historical results. The fluid activation **187** itself includes a variety of approaches. A first approach includes opening of either of the piston bypass **38-1** and piston bypass **38-2** allow the STF to move between the back channel **24** and the front channel **26** to lower the shear rate and thus select a lower viscosity which in turn allows more rapid movement of the piston in the chamber and hence speeds up the door. A second fluid activation approach includes opening of the chamber bypass **40** to lower the viscosity the STF. A third approach includes activating the set of emitters to directly alter the viscosity of the STF in a desired fashion (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0226] Alternatively, or in addition to, the fifth step further includes the computing entity **20-1** interpreting the output of the secondary object sensor to further update the updated object pattern of the plurality of proximal objects to produce a further updated object pattern of the plurality of proximal objects (e.g., continued to track the group of people walking about). Having produced the further updated object pattern, the fifth step further includes the computing entity **20-1** updating the fluid activation for the head unit device based on the further updated object pattern of the plurality of proximal objects and one or more of the piston velocity and the piston position as previously discussed.

[0227] Having determined the fluid activation for the head unit, a sixth step of the example method of operation includes the activating the set of fluid manipulation emitters **114-1-1** and **114-1-2** in accordance with the fluid activation **187** to manipulate one of the first shear threshold effect associated with the first piston bypass **38-1** and the second shear threshold effect associated with the second piston bypass **38-2** to control the motion of the object **12-1** (e.g., the first object) to control the motion of the object with regards to the second object.

[0228] The activating the set of fluid manipulation emitters in accordance with the fluid activation to control the motion of the first object with regards to the second object includes a variety of approaches. A first approach includes, when the piston is traveling through the chamber in the inward direction and when the STF is to have the decreasing viscosity, issuing the fluid activation to the set of fluid manipulation emitters to cause one of: the first piston bypass to facilitate the first shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF to facilitate the first range of shear rates.

[0229] Otherwise, when the STF is to have the increasing viscosity, the first approach includes issuing the fluid activation to the set of fluid manipulation emitters to cause one of: the first piston bypass to facilitate the first shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF to facilitate the second range of shear rates.

[0230] A second approach includes, when the piston is traveling through the chamber in the outward direction and when the STF is to have the decreasing viscosity, issuing the fluid activation to the set of fluid manipulation emitters to cause one of: the second piston bypass to facilitate the second shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF to facilitate the first range of shear rates.

[0231] Otherwise, when the STF is to have the increasing viscosity, the second approach includes issuing the fluid activation to the set of fluid manipulation emitters to cause one of: the second piston bypass to facilitate the second shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF to facilitate the second range of shear rates.

[0232] As another example, when the object **12-1** includes a large garage door and is opening when moving in the outward direction, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-2** to facilitate further opening of a one-way check valve to allow more of the STF to move from the front channel **26** to the back channel **24** thusly selecting the first range of shear rates and a lower viscosity of the STF to speed up the door to open when the detected set of secondary objects is a large vehicle passing through the large garage door when the historical results indicate that it is favorable to open the door quickly when the vehicle is very large.

[0233] As yet another example, when the set of secondary objects **12-2** through **12**-N includes a group of people, the computing entity **20-1** outputs the fluid activation **234-1-1** to the piston bypass **38-2** to facilitate closing down the one-way check valve to prevent STF from moving from the front channel **26** to the back channel **24** thusly selecting the second range of shear rates and a higher viscosity the STF to slow down a heavy door to close safely moving from the point Y**3** to the point Y**4** (e.g., when the piston is moving in the outward direction) as illustrated in FIG. **9**B when the historical results indicate that it is favorable to slow down the closing of the heavy door.

[0234] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. **1** can alternatively be performed by other modules of the system of FIG. **1** or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system **10**, cause one or more computing devices of the mechanical and computing system of FIG. **1** to perform any or all of the method steps described above.

[0235] FIGS. **10**A-**10**D are schematic block diagrams of an embodiment of a mechanical system to control an object. The mechanical system provides a head unit system that includes a head unit **10-2** and the object **12-1** of FIG. **1** (e.g., a first door, interchangeable referred to as a first object). In an embodiment, a computing system is operably coupled to the mechanical system, where the computing system includes one or more of the sensor **116-1-1** of FIG. **1**A, the emitter **114-1-1** of FIG. **1**A, the computing entity **20-1** of FIG. **1**A, and the computing entity **22** of FIG. **1**A.
[0236] The head unit **10-2** includes a shear thickening fluid (STF) **42**, where the STF **42** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.
[0237] The STF **42** includes nanoparticles. The nanoparticles includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO2, polystyrene, polymethylmethacrylate, or a mixture. The STF **42** further includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture.
[0238] The head unit **10-2** further includes a chamber **16** which includes two pieces that telescope. The chamber **16** is configured to contain a portion of the STF. The chamber **16** includes a front channel **26** and a back channel **24**. The chamber **16** is further configured to volumetrically expand and contract (e.g., via telescoping) to provide the first and second ranges of shear rates for the STF.
[0239] The head unit **10-2** further includes a gate **41**. The gate **41** is configured to separate the front channel **26** and the back channel **24** to control velocity of flow of the STF **42** between the front channel **26** and the back channel **24** by covering and uncovering an opening within the chamber **16** between the front channel **26** and the back channel **24**. The gate **41** is further configured to enable the STF **42** to provide the first range of shear rates as the STF flows from the back channel **24** to the front channel **26** when the gate **41** is in an open position uncovering the opening as the chamber **16** volumetrically expands (e.g., telescopes outward). The gate **41** is further configured to enable the STF **42** to provide the second range of shear rates as the STF **42** flows from the front channel **26** to the back channel **24** when the gate **41** is in a closed position covering the opening as the chamber **16** volumetrically contracts (e.g., telescopes inward).
[0240] The head unit **10-2** further includes a piston **36** housed at least partially radially within the back channel **24** of the chamber **16**. The piston **36** configured to load a spring **45** within the back channel **24** of the chamber **16** in response to the STF **42** flowing from the front channel **26** to back channel **24** when the chamber volumetrically contracts in response to a force applied to the chamber **16** from the object **12-1**. The spring **45** pushes up against a piston backstop **43** at the end of the chamber **16**.
[0241] The piston **36** is further configured to exert force on the STF **42** to cause the STF **42** to flow from the back channel **24** to front channel **26** in response to energy from the spring **45** unloading when the chamber **16** volumetrically expands absent the force applied to the chamber **16** from the object **12-1** (e.g., either the object **12-1** pulls the chamber outward as the object **12-1** moves away from the head unit **10-2** or the object **12-1** simply moves away from the chamber and the spring pushes the piston and the STF which telescopes the chamber outward).
[0242] The gate **41** further includes a set of bypass openings **39-1** and **39-2** between opposite sides of the gate **41**. The set of bypass openings are configured to enable the STF **42** to provide the second range of shear rates as the STF **42** flows from the front channel **26** through the set of bypass openings **39-1** and/or **39-2** to the back channel **24** when the gate **41** is in the closed position covering the opening as the chamber **16** volumetrically contracts. In an embodiment, there are any number of openings of a set of openings.
[0243] The gate **41** further includes a hinge **49**. The hinge **49** is configured to enable the gate **41** to swing to the open position from the closed position and to swing to the closed position from the open position. In an embodiment, a stop component of the hinge prevents the hinge from opening a full 90 degrees. For example, the stop component holds the gate open at a 45 degree maximum to

help prevent the gate from sticking in the full open position.

[0244] The set of bypass openings **39-1** and **39-2** further includes one or more of a variety of configurations. A first configuration includes a first bypass opening configured with at least one cylindrical tube with substantially consistent diameter from one side of the gate to an opposite side of the gate. For example, a straight tube to enable even STF flow.

[0245] A second configuration includes a second bypass opening configured with at least one conical shaped tube with an increasing diameter from the one side of the gate to the opposite side of the gate. For example, a tapered tube.

[0246] A third configuration includes a third bypass opening configured with the at least one conical shaped tube with a decreasing diameter from the one side of the gate to the opposite side of the gate. For example, the tapered tube in an opposite direction.

[0247] A fourth configuration includes a fourth bypass opening configured with at least one venturi shaped tube from the one side of the gate to the opposite side of the gate. For example, the venturi to cause a venturi effect between sides of the gate.

[0248] In an embodiment, the head unit **10-2** further includes a chamber bypass **40** (e.g., as discussed with reference to FIG. **5**A) between opposite ends of the chamber **16**. The chamber bypass **40** facilitates flow of a portion of the STF between the opposite ends of the chamber **16** to facilitate equalization of pressure of the STF when the piston **36** travels through the chamber **16** in an inward or an outward direction.

[0249] In a further embodiment, the head unit system further includes the set of fluid flow sensors **116-1-1** and **116-1-2** positioned proximal to the chamber **16** as discussed with reference to FIG. **5**A. The set of fluid flow sensors are configured to provide a fluid response from the STF **42**. The set of fluid flow sensors includes one or more of a valve opening detector associated with a set of bypass openings, a mechanical position sensor, an image sensor, and a light sensor. The fluid flow sensors further includes an audio sensor, a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.

[0250] In a still further embodiment, the head unit system further includes the set of fluid manipulation emitters **114-1-1** and **114-1-2** positioned proximal to the chamber as discussed with reference to FIG. **5**A. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF **42**, the gate **41**, the piston **36**, and the set of bypass openings **39-1** and **39-2** to provide the control of the motion of the object **12-1**. The set of fluid manipulation emitters includes one or more of a variable flow valve associated with the set of bypass openings, a mechanical vibration generator, an image generator, a light emitter, an audio transducer, and a speaker. The set of fluid manipulation emitters further includes an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter.

[0251] FIG. **10**A illustrates an example of operation of the head unit system where, in a first step, the head unit system is at idle. The object **12-1** is operably coupled to the chamber **16** but, the first step, exerts no force upon the chamber **16**. The chamber **16** is telescoped to a full outward position and the STF **42** is contained within the chamber in both the front channel **26** and the back channel **24** up to the piston **36**.

[0252] FIG. **10**B further illustrates the example of operation of the head unit system where, in a second step, the head unit system moves from the idle position to a slow movement due to STF resistance. The gate **41** starts in the closed position such that when force is applied from the object **12-1** to the chamber **16** to compress the chamber **16**, the STF **42** in the front channel **26** runs up against the closed gate **41** causing higher shear forces to trigger the second range of shear rates. Some of the STF flows through the bypass **39-1** and the bypass **39-2** from the front channel **26** to the back channel **24**.

[0253] The bypass openings further include one or more of a one-way check valve and a variable flow valve. When the piston **36** is traveling through the chamber **16** in the inward direction away from the gate **41**, a first setting of the variable flow valve facilitates the second range of shear rates when the STF is to have the increasing viscosity.

[0254] As the STF fills the back channel **24** the piston **36** is pushed deeper into the chamber **16** causing loading of the spring **45** up against the piston backstop **43**. When the piston **36** is traveling through the chamber **16** in an inward direction away from the gate **41**, a second shear threshold effect includes the second range of shear rates when the STF **42** is configured to have the increasing viscosity. As such, the FIG. **10**C further illustrates the example of operation of the head unit system where, in a third step, the head unit system moves from the slow movement due to STF resistance position (e.g., compressed chamber) to a fast from open gate position (e.g., the object **12-1** moves back towards a starting position). As the object **12-1** moves outward (e.g., away from the chamber **16** such that the chamber **16** expands or the object **12-1** pulls the chamber **16**) the STF in the back channel **24** is pushed up against the gate **41** by the piston **36** due to the spring **45** releasing its stored energy. The gate **41** opens in response to the STF **42** moving towards the gate **41** due to the pressure exerted on the STF **42** by the piston **36**.

[0255] As the gate **41** opens, the STF moves from the back channel **24** to the front channel **26** with a higher velocity as compared to the opposite direction since the STF **42** is both pushed by the piston **36** and pulled by the suction of the STF **42** within the front channel **26** as the front channel volumetrically expands as the chamber expands. When the one-way check valve is utilized and the piston **36** is traveling through the chamber **16** in the outward direction towards the gate **41**, the one-way check valve is configured to prevent STF flow through the one-way check valve. The STF **42** flows through the open gate into the front channel **26**.

[0256] When the piston **36** is traveling through the chamber **16** in an outward direction towards the gate **41**, a first shear threshold effect includes the first range of shear rates when the STF **42** is configured to have the decreasing viscosity. As such, a lower viscosity associated with this chamber expansion enables expansion of the chamber **16** at a faster rate than the compression of the chamber **16** as previously discussed. In an embodiment, the hinge **49** is configured to limit the opening of the gate **41** to an angle between 0° and 90°. For example, the hinge **49** is configured to limit the opening of the gate **41** to 45° such that the gate **41** is less likely to not remain stuck open when it is time to compress the chamber **16**.

[0257] FIG. **10**D further illustrates the example of operation of the head unit system where, in a fourth step, the head unit system moves from the fast from open gate position to the stopped and idle position. The spring **45** having released all its energy, positions the piston **36** at a steady state position with regards to the force the spring applies up against the STF **42**. The STF **42** stabilizes between the back channel **24** and the front channel **26** when the chamber **16** has reached its maximum expansion level. In response, the gate **41** tends towards closing over the opening between the back channel **24** and the front channel **26**. In an embodiment, the hinge **49** includes a return spring to force the gate **41** closed when the pressure of the back channel **24** and the front channel **26** is substantially the same.

[0258] In another embodiment, when the sensors and emitters are utilized in conjunction with a computing entity, the rate at which the second shear threshold effect takes place is sensed and automatically adjusted for a desired rate such that the object **12-1** travels inward at a desired trajectory (e.g., a maximum speed, a maximum acceleration, a desired length of travel, etc.). In a similar fashion, the sensors and emitters in conjunction with the computing entity are utilized to automatically adjust for another desired rate with regards to outward travel of the object **12-1**.

[0259] FIGS. **11**A-**11**D are schematic block diagrams of another embodiment of a mechanical system to control an object. The mechanical system provides a head unit system that includes a head unit **10-3** and the object **12-1** of FIG. **1** (e.g., a first door, interchangeable referred to as a first object). In an embodiment, a computing system is operably coupled to the mechanical system, where the computing system includes one or more of the sensor **116-1-1** of FIG. **1**A, the emitter **114-1-1** of FIG. **1**A, the computing entity **20-1** of FIG. **1**A, and the computing entity **22** of FIG. **1**A.

[0260] The head unit **10-3** includes a shear thickening fluid (STF) **42**, where the STF **42** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.
[0261] The STF **42** includes nanoparticles. The nanoparticles include one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO2, polystyrene, polymethylmethacrylate, or a mixture. The STF **42** further includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture.
[0262] The head unit **10-3** further includes a chamber **16**. The chamber **16** is configured to contain a portion of the STF **42**. The chamber **16** includes a front channel **26** and a back channel **24**.
[0263] The head unit **10-3** further includes a set of gates **57-1** and **57-2**. Any number of gates may be utilized. The set of gates is configured to separate the front channel **26** and the back channel **24** to control velocity of flow of the STF **42** between the front channel **26** and the back channel **24**. The set of gates **57-1** and **57-2** is further configured to enable the STF **42** to provide the first range of shear rates as the STF flows from the back channel **24** to the front channel **26** when the set of gates is in an open position. The set of gates **57-1** and **57-2** is further configured to enable the STF **42** to provide the second range of shear rates as the STF **42** flows from the front channel **26** to the back channel **24** when the set of gates is in a closed position. In an embodiment, the set of gates open and close in unison (e.g., closed together or open substantially the same amount simultaneously).
[0264] The set of gates **57-1** and **57-2** further includes a set of hinges **59-1** and **59-2**. The set of hinges is configured to enable the set of gates **57-1** and **57-2** to swing to the open position from the closed position and to swing to the closed position from the open position. In an embodiment, the set of hinges is further configured to only allow the set of gates to open to a maximum open level. For example, the set of hinges only allow the gates to open to a maximum of 45° within a range of zero to 90°.
[0265] The head unit **10-3** further includes a cupped piston **55**. The cupped piston is configured to facilitate flow of the STF **42** through a slot set **63** between the front channel **26** and the back channel **24**. The slot set includes one or more varieties of slots. A first variety of the slot set is configured with at least one cylindrical tube with substantially consistent diameter from one side of the cupped piston to an opposite side of the cupped piston. For example, a uniform hole.
[0266] A second variety of slot of the slot set is configured with at least one conical shaped tube with an increasing diameter from the one side of the cupped piston to the opposite side of the cupped piston. For example, a tapered hole.
[0267] A third variety of slot of the slot set is configured with the at least one conical shaped tube with a decreasing diameter from the one side of the cupped piston to the opposite side of the cupped piston. For example, another tapered hole in another direction.
[0268] A fourth variety of slot of the slot set is configured with at least one venturi shaped tube from the one side of the cupped piston to the opposite side of the cupped piston. For example, the venturi shaped tube to produce a venturi effect between opposite sides of the cupped piston.
[0269] The head unit **10-3** further includes a bushing **53**. The bushing is configured within the chamber **16** to cause containment of the STF **42** in the back channel **24**. In an embodiment, the bushing **53** is further configured with one or more seals **47** provide further containment of the STF **42** within the back channel **24**. The head unit **10-3** further includes a backstop **51**. The backstop is configured with the chamber **16** to cause containment of the STF **42** in the front channel **26**.
[0270] The head unit **10-3** further includes a plunger **28** housed at least partially radially within the back channel **24** of the chamber **16**. The plunger is configured to facilitate the set of gates to operate in the closed position to cause the STF **42** within the front channel **26** to provide the second range of shear rates in response to an inward force applied to the plunger **28** from the object **12-1** as the STF **42** flows from the front channel **26** to the back channel **24**. The plunger **28** is further

configured to facilitate the set of gates **57-1** and **57-2** to operate in the open position to cause the STF **42** within the back channel **24** to provide the first range of shear rates in response to an outward force applied to the plunger as the STF **42** flows from the back channel **24** to the front channel **26**.

[0271] The head unit **10-3** further includes a cap **61**. The cap is configured to move in unison with the plunger **28** radially within the chamber **16**. In an embodiment, the **61** is operably coupled to the plunger **28** such that a force applied by the object **12-1** to the plunger **28** and the **61** causes the **61** and the plunger **28** to move in unison in the inward direction in the chamber **16**. The head unit **10-3** further includes a spring **45**. The spring **45** is configured to store energy when the **61** moves in the inward direction in response to a force from the object **12-1**. The spring **45** is further configured to release the energy, absent the force applied by the object **12-1**, such that the **61** and plunger **28** move in the outward direction quickly to reset the **61** to the idle position.

[0272] In an embodiment, the head unit **10-3** further includes a chamber bypass **40** as discussed with reference to FIG. **1**A between opposite ends of the chamber **16**. The chamber bypass **40** facilitates flow of a portion of the STF **42** between the opposite ends of the chamber **16** when the set of gates **57-1** and **57-2** travels through the chamber **16** in an inward or an outward direction.

[0273] In an embodiment where the head unit system further includes the set of fluid flow sensors, the set of fluid flow sensors are configured to provide a fluid response from the STF **42**. The set of fluid flow sensors includes one or more of a valve opening detector associated with the slot set **63** of the cupped piston **55**, a mechanical position sensor, an image sensor, a light sensor, and an audio sensor. The set of fluid flow sensors further includes a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.

[0274] In an embodiment where the head unit system further includes the set of fluid manipulation emitters, the set of fluid manipulation emitters are positioned proximal to the chamber **16**. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF **42**, the set of gates **57-1** and **57-2**, and the slot set **63** of the cupped piston **55** to provide the control of the motion of the object **12-1**. The set of fluid manipulation emitters includes a variety of fluid manipulation emitters including one or more of a variable flow valve associated with the slot set **63**, a mechanical vibration generator, an image generator, a light emitter, and an audio transducer. The set of fluid manipulation emitters further includes a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter.

[0275] FIG. **11**A illustrates an example of operation of the head unit system where, in a first step, the head unit system is in an idle position. The object **12-1** is operably coupled to the plunger **28** and cap **61** ready to exert force upon the plunger and. At idle, the plunger and cap are at a full outward position such that the set of gates are in the closed position and are ready to exert force upon the STF **42** within the front channel **26**.

[0276] FIG. **11**B further illustrates the example of operation of the head unit system where, in a second step, the head unit system moves from the idle position to a slow compression operation due to STF resistance. For example, the object **12-1** applies the force to the cap **61** and the plunger **28** such that the set of gates **57-1** and **57-2** travels through the chamber **16** in the inward direction away from the back channel **24** to produce a second shear threshold effect as the STF **42** within the front channel **26** is compressed by the closed set of gates and the cupped piston. The second shear threshold effect includes the second range of shear rates when the STF is configured to have the increasing viscosity.

[0277] The set of gates **57-1** and **57-2** further include a bypass opening set **65**. The bypass opening set **65** is configured to facilitate flow of the STF **42** from the front channel **26** through the bypass opening set **65** to the back channel **24** to cause the STF **42** within the front channel **26** to provide the second range of shear rates when the set of gates **57-1** and **57-2** is operating in the closed position. Movement of the set of gates and the cupped piston towards the front channel **26** in response to the force applied from the plunger **28** serves to facilitate closure of the set of gates. As the set of gates closes, the STF **42** flows from the front channel **26** to the back channel **24** by one or more approaches. A first approach includes traveling around the set of gates when a diameter of the set of gates is less than a diameter of the chamber **16**. A second approach includes traveling through the set of gates via one or more bypass openings of the bypass opening set **65**.

[0278] The cupped piston **55** is further configured to facilitate flow of the STF **42** around the set of gates between the chamber and the set of gates from the front channel through the slot set **63** of the cupped piston to the back channel to cause the STF within the front channel to provide the second range of shear rates when the set of gates is operating in the closed position. For example, a smaller diameter slot set facilitates higher shear thresholds and a higher viscosity from the second range of shear rates such that a relatively small force from the object **12-1** causes an abrupt slow down of the movement of the object **12-1**.

[0279] The cupped piston **55** is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the front channel through the slot set of the cupped piston to the back channel to cause the STF within the front channel to provide a sub-range of the second range of shear rates in accordance with a slot size configuration of the slot set when the set of gates is operating in the closed position. For example, a larger diameter slot set requires even more force from the object **12-1** to reach the second range of shear rates to cause the slowing down of the movement of the object **12-1**.

[0280] The second step of the example of operation of the head unit **10-3** ends with the cap **61** and the plunger **28** reaching an end of travel (e.g., the cap **61** travels inward within the chamber **16** to rest when reaching the bushing **53**). Soon after reaching the end of travel, the pressure of the STF within the front channel **26** equalizes with the pressure of the STF within the back channel **24**. As long as a maintaining force from the object **12-1** is applied to the plunger **28** the head unit **10-3** remains in this position with resistance to the force from the object **12-1** coming from this stored energy of the spring **45** and no longer from the STF **42**.

[0281] FIG. **11**C further illustrates the example of operation of the head unit system where, in a third step, the head unit system moves from the slow compression operation to a fast rebounding operation absent the force applied from the object **12-1**. The set of gates **57-1** and **57-2** is further configured such that, when the set of gates is traveling through the chamber in the outward direction towards the back channel, a first shear threshold effect is produced. The first shear threshold effect includes the first range of shear rates when the STF is configured to have the decreasing viscosity.

[0282] As the force from the object **12-1** is removed from the plunger **28** and the cap **61** or when the object **12-1** pulls the plunger **28** in the outward direction, the spring **45** is further configured within the chamber to have stored the energy as a result of the inward force applied to the plunger and to the cap from the object **12-1**. The spring **45** is further configured to move the plunger **28**, the cap **61**, and set of gates **57-1** and **57-2** outward absent the inward force applied to the plunger and to the cap from the object **12-1**. As the set of gates is pulled by the plunger from the front channel to the back channel, the force of the STF serves to open the set of gates.

[0283] The cupped piston **55** is further configured to facilitate the flow of the STF from the back channel **24** through the slot set **63** of the cupped piston **55** to the front channel to cause the set of gates to operate in the open position and to cause the STF within the back channel to provide the first range of shear rates when the set of gates is operating in the open position. As such, an improvement to the operation of the head unit system is provided where the cap **61** quickly rebounds to the idle position absent the force applied by the object **12-1**.

[0284] FIG. **11**D further illustrates the example of operation of the head unit system where, in a fourth step, the head unit system transitions from the fast rebounding operation to a stopping operation where the system finally comes to a steady-state rest absent the force applied by the object **12-1**. For example, the spring **45** releases all of its energy against the cap **61** and plunger **28** thus pulling the cupped piston **55** in the set of gates **57-1** and **57-2** to a fully outward resting position. The gates **57-1** and **57-2** begin to close when the pressure of the STF equalizes between the back channel **24** and the front channel **26**. As such, the head unit **10-3** is resetting to be ready for another cycle where the object **12-1** applies the force to the head unit **10-3**.

[0285] FIGS. **12**A-**12**F are schematic block diagrams of another embodiment of a mechanical system to control an object. The mechanical system provides a head unit system that includes a head unit **10-4** and the objects **12-2** and **12-3** of FIG. **1**A. In an embodiment, a computing system is operably coupled to the mechanical system, where the computing system includes one or more of the sensor **116-1-1** of FIG. **1**A, the emitter **114-1-1** of FIG. **1**A, the computing entity **20-1** of FIG. **1**A, and the computing entity **22** of FIG. **1**A.

[0286] FIGS. **12**A and **12**B illustrates an example of operation of the mechanical system to control the object. In an embodiment, the object **12-2** includes a first portion of a hinge assembly that is associated with a doorjamb and the object **12-2** includes a remaining portion of the hinge assembly that is associated with a door that swings utilizing the hinge assembly. In particular, the object **12-3** includes both the remaining portion of the hinge assembly and the door and the object **12-2** includes the first portion of the hinge assembly and the doorjamb.

[0287] FIG. **12**A illustrates a relationship between the objects **12-2** and **12-3** with regards to the head unit **10-4** where the hinge assembly is in the open position. FIG. **12**B further illustrates the relationship between objects **12-2** and **12-3** with regards to the head unit **10-4** where the hinge assembly is rotating towards a closed position (e.g., as the door closes). As will be discussed below, rotation of the object **12-3** causes a portion of the head unit **10-4** to rotate which causes compression of a shear thickening fluid within the head unit **10-4** to achieve a desired viscosity of the shear thickening fluid which in turn limits acceleration and/or velocity of the closing of the door.

[0288] FIG. **12**C illustrates an embodiment of the head unit **10-4** and a first step of an example of operation of the head unit system that includes the head unit **10-4** at a starting open position. The head unit **10-4** includes a shear thickening fluid (STF) **42**, where the STF **42** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0289] The STF **42** includes nanoparticles. The nanoparticles include one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO2, polystyrene, polymethylmethacrylate, or a mixture. The STF **42** further includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture.

[0290] The head unit **10-4** further includes a chamber **75**. The chamber **75** is configured to contain a portion of the STF **42**. The chamber **75** includes a front channel **26** and a back channel **24**. In an embodiment, the chamber **75** is associated with the object **12-2** (e.g., the doorjamb) and is considered a fixed portion of the head unit **10-4**. When the chamber **75** is considered the fixed portion, a rotating portion of the head unit **10-4** includes one or more rotating chambers **73-1** and **73-2**. The one or more rotating chambers are associated with the object **12-3** (e.g., the door) and are considered a moving and/or rotational portion of the head unit **10-4** relative to the chamber **75** and the object **12-2**.

[0291] The head unit **10-4** further includes a set of gates **57-1** and **57-2**. Any number of gates may be utilized. The set of gates is configured to separate the front channel **26** and the back channel **24** to control velocity of flow of the STF **42** between the front channel **26** and the back channel **24**. The set of gates **57-1** and **57-2** is further configured to enable the STF **42** to provide the first range of shear rates as the STF flows from the back channel **24** to the front channel **26** when the set of gates is in an open position. The set of gates **57-1** and **57-2** is further configured to enable the STF **42** to provide the second range of shear rates as the STF **42** flows from the front channel **26** to the back channel **24** when the set of gates is in a closed position. In an embodiment, the set of gates open and close in unison (e.g., closed together or open substantially the same amount simultaneously). In another embodiment, the set of gates open within a first time frame and close within a second time frame. In yet another embodiment, the set of gates open sequentially. In a still further embodiment, the set of gates close sequentially.

[0292] The set of gates **57-1** and **57-2** further includes a set of hinges **59-1** and **59-2**. The set of hinges is configured to enable the set of gates **57-1** and **57-2** to swing to the open position from the closed position and to swing to the closed position from the open position. In an embodiment, the set of hinges is further configured to only allow the set of gates to open to a maximum open level. For example, the set of hinges only allow the gates to open to a maximum of 45° within a range of zero to 90°.

[0293] The set of gates **57-1** and **57-2** further include a bypass opening set **65**. The bypass opening set **65** is configured to facilitate flow of the STF **42** from the front channel **26** through the bypass opening set **65** to the back channel **24** to cause the STF **42** within the front channel **26** to provide the second range of shear rates when the set of gates **57-1** and **57-2** is operating in the closed position. Movement of the set of gates towards the front channel **26** in response to a force applied from the plunger **28** serves to facilitate closure of the set of gates. As the set of gates closes, the STF **42** flows from the front channel **26** to the back channel **24** by one or more approaches. A first approach includes traveling around the set of gates when a diameter of the set of gates is less than a diameter of the chamber **75**. A second approach includes traveling through the set of gates via one or more bypass openings of the bypass opening set **65**.

[0294] The head unit **10-4** further includes a plunger **28** housed at least partially radially within the back channel **24** of the chamber **75**. The plunger is configured to facilitate the set of gates to operate in the closed position to cause the STF **42** within the front channel **26** to provide the second range of shear rates inward force applied to the plunger **28** from a rotational force of the object **12-3** as the STF **42** flows from the front channel **26** to the back channel **24**. The plunger **28** is further configured to facilitate the set of gates **57-1** and **57-2** to operate in the open position to cause the STF **42** within the back channel **24** to provide the first range of shear rates in response to an outward force applied to the plunger **28** from a counter-rotational force of the object **12-3** as the STF **42** flows from the back channel **24** to the front channel **26**.

[0295] The head unit **10-4** further includes a threaded plunger **79**. The threaded plunger **79** is configured to apply the inward force to the plunger **28** in response to the rotational force of the object **12-3**. The threaded plunger **79** may also be referred to as a threaded non-rotational plunger **79**. The threaded plunger **79** is further configured to generate the inward force when a nut **81** rotates on threads of the threaded plunger **79** of as a result of the rotational force of the object **12-3**. In an embodiment, the nut **81** is configured within a rotating chamber **73-2** that is extended beyond the chamber **75**. The nut **81** is configured in a fixed position within the rotating chamber **73-2** by way of a fastener **83** between the rotating chamber **73-2** and the nut **81**.

[0296] The threaded plunger **79** is further configured to apply the outward force to the plunger **28** in response to the counter-rotational force of the object **12-3**. The threaded plunger **79** is further configured to generate the outward force when the nut **81** counter-rotates on the threads of the threaded plunger **79** of as a result of the counter-rotational force of the object **12-3**.

[0297] In an embodiment, the head unit **10-4** further includes a stop adjuster **87**. The stop adjuster **87** is configured with the nut **81** to prevent the threaded plunger from traveling beyond a stop position of the stop adjuster **87**. As a result, the door can only swing open to a maximum open position.

[0298] The head unit **10-4** further includes a cupped piston **55**. The cupped piston is configured to facilitate flow of the STF **42** around the set of gates between the chamber **75** and the set of gates **57-1** and **57-2** from the front channel **26** through a slot set **63** of the cupped piston **55** to the back channel **24** to cause the STF **42** within the front channel **26** to provide the second range of shear rates when the set of gates is operating in the closed position. The cupped piston **55** is further configured to facilitate flow of the STF **42** from the back channel **24** through the slot set **63** of the cupped piston **55** to the front channel **26** to cause the set of gates to operate in the open position and to cause the STF **42** within the back channel **24** to provide the first range of shear rates when the set of gates is operating in the open position.

[0299] The cupped piston is further configured to facilitate flow of the STF **42** facilitate flow of the STF through the slot set **63** between the front channel **26** and the back channel **24** utilizing one or more varieties of slots. A first variety of the slot set is configured with at least one cylindrical tube with substantially consistent diameter from one side of the cupped piston to an opposite side of the cupped piston. For example, a uniform hole.

[0300] A second variety of slot of the slot set is configured with at least one conical shaped tube with an increasing diameter from the one side of the cupped piston to the opposite side of the cupped piston. For example, a tapered hole.

[0301] A third variety of slot of the slot set is configured with the at least one conical shaped tube with a decreasing diameter from the one side of the cupped piston to the opposite side of the cupped piston. For example, another tapered hole in another direction.

[0302] A fourth variety of slot of the slot set is configured with at least one venturi shaped tube from the one side of the cupped piston to the opposite side of the cupped piston. For example, the venturi shaped tube to produce a venturi effect between opposite sides of the cupped piston.

[0303] The head unit **10-4** further includes a fixed bushing **77**. The fixed bushing is configured in a fixed position relative to the chamber **75** to cause containment of the STF **42** in the back channel **24**. In an embodiment, the fixed bushing **77** is further configured with one or more seals **47** provide further containment of the STF **42** within the back channel **24**. In an embodiment, the fastener **83** stabilizes the fixed bushing **77** in the fixed position with regards to the chamber **75**.

[0304] In an embodiment, the fixed bushing **77** further includes a dowel key **85 2** guide the plunger **28** through the fixed bushing **77** as the plunger **28** travels longitudinally through the chamber **75** without rotation.

[0305] The head unit **10-4** further includes a backstop **71**. The backstop is configured with the chamber **75** to cause containment of the STF **42** in the front channel **26**. In an embodiment, the backstop **71** includes one or more seals **47** when the backstop **71** is configured within the chamber **75** to prevent flow of the STF into the rotating chamber **73-1**. The backstop **71** is secured to the rotating chamber **73-1** by way of another fastener **83**.

[0306] The head unit **10-4** further includes a piston **36**. The piston **36** is configured to travel longitudinally within the chamber **75** to cause containment of the STF **42** in the back channel **24**. The piston **36** is further configured to compress a spring **45** within the chamber **75** between the piston **36** and the fixed bushing **77** relative to the chamber **75** when the STF **42** flows from the front channel **26** to the back channel **24** in response to the inward force to the plunger **28**. The piston **36** is further configured to cause flow of the STF **42** in the back channel **24** in response to de-compression of the spring **45** further causing the set of gates **57-1** and **57-2** to operate in the open position and to further cause the STF **42** within the back channel **24** to provide the first range of shear rates when the set of gates is operating in the open position when the outward force is applied to the plunger **28**.

[0307] In an embodiment, the head unit **10-4** further includes a chamber bypass **40** as discussed with reference to FIG. **1**A between opposite ends of the chamber **75**. The chamber bypass **40** facilitates flow of a portion of the STF **42** between the opposite ends of the chamber **75** when the set of gates **57-1** and **57-2** travels through the chamber **75** in an inward or an outward direction.

[0308] In an embodiment where the head unit system further includes the set of fluid flow sensors, the set of fluid flow sensors are configured to provide a fluid response from the STF **42**. The set of fluid flow sensors includes one or more of a valve opening detector associated with the slot set **63** of the cupped piston **55**, a mechanical position sensor, an image sensor, a light sensor, and an audio sensor. The set of fluid flow sensors further includes a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.

[0309] In an embodiment where the head unit system further includes the set of fluid manipulation emitters, the set of fluid manipulation emitters are positioned proximal to the chamber **75**. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF **42**, the set of gates **57-1** and **57-2**, and the slot set **63** of the cupped piston **55** to provide the control of the motion of the object **12-3**. The set of fluid manipulation emitters includes a variety of fluid manipulation emitters including one or more of a variable flow valve associated with the slot set **63**, a mechanical vibration generator, an image generator, a light emitter, and an audio transducer. The set of fluid manipulation emitters further includes a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter.

[0310] FIG. **12**C further illustrates an example of operation of the head unit system where, in a first step, the head unit system is in an open position. The object **12-3** is operably coupled to the rotating chamber **73-2** ready to exert the rotational force upon the rotating chamber **73-2**. At idle, the threaded non-rotational plunger **79** and the plunger **28** plunger are at a full outward position such that the set of gates are in the closed position and are ready to exert force upon the STF **42** within the front channel **26**. At idle, in the open position, the spring **45** is relaxed and the piston **36** is at a steady state with regards to spring pressure from the spring **45** and pressure of the STF within the back channel **24**.

[0311] FIG. **12**D further illustrates the example of operation of the head unit system where, in a second step, the head unit system moves from the open position to a rotational operation due to the rotational force from the object **12-3** applied to the rotating chamber **73-2**. For example, the object **12-3** applies the rotational force to the rotating chamber **73-2** which turns the nut **81** causing the threaded non-rotation plunger **79 2** travel inward through the fixed bushing **77** exerting force on the operably coupled plunger **28** causing the plunger **28** to exert force on the cupped piston **55** and set of gates **57-1** and **57-2**. When set of gates **57-1** and **57-2** travels through the chamber **75** in the inward direction away from the back channel **24** that produces a second shear threshold effect as the STF **42** within the front channel **26** is compressed by the closed set of gates and the cupped piston **55**. The second shear threshold effect includes the second range of shear rates when the STF is configured to have the increasing viscosity.

[0312] The cupped piston **55** is further configured to facilitate flow of the STF **42** around the set of gates between the chamber and the set of gates from the front channel through the slot set **63** of the cupped piston to the back channel to cause the STF within the front channel to provide the second range of shear rates when the set of gates is operating in the closed position. For example, a smaller diameter slot set facilitates higher shear thresholds and a higher viscosity from the second range of shear rates such that a relatively small force from the object **12-3** causes an abrupt slow down of the movement of the object **12-3** (e.g., stops the door from slamming the door jamb).

[0313] The cupped piston **55** is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the front channel through the slot set of the cupped piston to the back channel to cause the STF within the front channel to provide a sub-range of the second range of shear rates in accordance with a slot size configuration of the slot set when the set of gates is operating in the closed position. For example, a larger diameter slot set requires even more rotational force from the object **12-3** to reach the second range of shear rates to cause the slowing down of the movement of the object **12-3**.

[0314] The second step of the example of operation of the head unit **10-4** ends with the nut **81** causing an end of travel of the threaded non-rotational plunger **79**. Soon after reaching the end of travel, the pressure of the STF within the front channel **26** equalizes with the pressure of the STF within the back channel **24**. As long as a maintaining force from the object **12-3** (e.g., the doors closed and latched by a doorknob) is applied to the plunger **28** the head unit **10-4** remains in this position with resistance to the force from the object **12-3** coming from this stored energy of the spring **45** and no longer from the STF **42**.

[0315] FIG. **12**E further illustrates the example of operation of the head unit system where, in a third step, the head unit system moves from the rotation operation (e.g., and idling with the cupped piston **55** fully extended within the front channel **26**) to a counter-rotation operation as counter-rotation force applied from the object **12-3** (e.g., the doors opened). The set of gates **57-1** and **57-2** is further configured such that, when the set of gates is traveling through the chamber in the outward direction towards the back channel, a first shear threshold effect is produced. The first shear threshold effect includes the first range of shear rates when the STF is configured to have the decreasing viscosity.

[0316] As the counter-rotational force from the object **12-3** counter-rotates the nut **81** causing the threaded non-rotational plunger **79** to move outward within the rotating chamber **73-2** thus pulling the plunger **28** and the cupped piston **55** along with the gates **57-1** and **57-2** in the outward direction, the spring **45** is further configured within the chamber **75** to release the stored energy as a result of the STF **42** flowing from the back channel **24** through the cupped piston **55** and the open gates **57-1** and **57-2** into the front channel **26**. The spring **45** is further configured to assist moving the STF **42** from the back channel **24** to the front channel **26** as the cupped piston **55** is drawn outward by the counter-rotation movement. As the set of gates **57-1** and **57-2** is pulled inward by the plunger **28** from the front channel **26** to the back channel **24**, the force of the STF **42** serves to open the set of gates **57-1** and **57-2** causing even more STF **42** to flow through the cupped piston **55** with less resistance thus speeding up the counter-rotation movement of the object **12-3** (e.g., the door opens quickly.

[0317] The cupped piston **55** is further configured to facilitate the flow of the STF from the back channel **24** through the slot set **63** of the cupped piston **55** to the front channel **26** to cause the set of gates **57-1** and **57-2** to operate in the open position and to cause the STF **42** within the back channel to provide the first range of shear rates when the set of gates is operating in the open position. As such, an improvement to the operation of the head unit system is provided where the rotating chamber **73-2** quickly rebounds enabling the object **12-3** to counter-rotate back to the open position.

[0318] FIG. **12**F further illustrates the example of operation of the head unit system where, in a fourth step, the head unit system transitions from the counter-rotation operation to a stopping operation where the system finally comes to a steady-state rest as the object **12-3** returns to the open an idle position. For example, the spring **45** releases all of its energy against the piston **36** thus pushing the STF **42** through the set of gates **57-1** and **57-2** to a fully inward resting position where pressure of the STF equalizes between the back channel **24** and the front channel **26**. The gates **57-1** and **57-2** begin to close when that pressure of the STF equalizes. As such, the head unit **10-4** is resetting to be ready for another cycle where the object **12-3** applies the rotational force to the rotating chamber **73-2**.

[0319] FIGS. **13**A-**13**D are schematic block diagrams of another embodiment of a mechanical head unit system to control an object. FIG. **13**A illustrates the head unit system to include a shear thickening fluid (STF) module **206** to be coupled and mutually operated with a hinge module **204**. To control motion of an object (e.g., a door coupled to the hinge module).

[0320] FIG. **13**B illustrates an embodiment of an STF hinge unit **202** that includes the STF module **206** coupled to the hinge module **204** utilizing a retaining ring **210**. The STF module **206** includes a shear thickening fluid (STF) **42**. The STF **42** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0321] The STF **42** includes a plurality of nanoparticles. The plurality of nanoparticles includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO2, polystyrene, polymethylmethacrylate, or a mixture thereof. The STF **42** further includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture thereof.

[0322] The STF module **206** further includes a chamber **16**. The chamber **16** is configured to

contain a portion of the STF **42**. The chamber **16** includes a front channel **26** and a back channel **24**.
[0323] In an embodiment, the STF module **206** further includes a set of gates **57-1** and **57-2**. Alternatively, in another embodiment, the set of gates are replaced by functionality of slots of a piston as discussed below. The set of gates is configured to separate the front channel **26** and the back channel **24** to control velocity of flow of the STF **42** between the front channel **26** and the back channel **24**.
[0324] The set of gates is further configured to enable the STF **42** to provide the first range of shear rates as the STF flows from the back channel **24** to the front channel **26** when the set of gates **57-1** and **57-2** is in an open position. The set of gates is further configured to enable the STF **42** to provide the second range of shear rates as the STF flows from the front channel **26** to the back channel **24** when the set of gates is in a closed position.
[0325] The STF module **206** further includes a piston (e.g., a cupped piston **55**). The piston is housed at least partially radially within the chamber **16**. The piston is configured to include a slot set. The piston is further configured to facilitate flow of the STF **42** through the slot set between the front channel **26** and the back channel **24** when the piston travels longitudinally within the chamber **16**.
[0326] The slot set includes one or more varieties of slots. A first slot of the slot set configured with at least one cylindrical tube with substantially consistent diameter from one side of the piston to an opposite side of the piston. A second slot of the slot set configured with at least one conical shaped tube with an increasing diameter from the one side of the piston to the opposite side of the piston. A third slot of the slot set configured with the at least one conical shaped tube with a decreasing diameter from the one side of the piston to the opposite side of the piston. A fourth slot of the slot set configured with at least one venturi shaped tube from the one side of the piston to the opposite side of the piston.
[0327] The STF module **206** further includes a plunger **28-1**. The plunger **28-1** is housed at least partially radially within the back channel **24** of the chamber **16**. The plunger **28-1** is configured to exert force on the piston in response to the motion of the object.
[0328] The STF module **206** further includes a bushing **53** and at least one seal **47** to provide a seal between the bushing **53** and the chamber **16**. The bushing **53** configured in a fixed position relative to the chamber **16** to cause containment of the STF **42** in the back channel **24**.
[0329] The STF module **206** further includes a backstop **51**. The backstop **51** is configured with the chamber **16** to cause containment of the STF **42** in the front channel **26**.
[0330] The STF module **206** further includes a cap **61**. The cap **61** is configured to move in unison with the plunger **28-1** radially within the chamber **16**.
[0331] The STF module **206** further includes a spring **45**. The spring **45** is configured to store energy when the cap **61** moves in an inward direction in response to a force from the object. The spring is further configured to release the energy, absent the force from the object, such that the cap **61** and the plunger **28-1** move in an outward direction.
[0332] The hinge module **204** includes a threaded plunger (a threaded non-rotational plunger **79**). The threaded plunger is configured to facilitate an inward force to the plunger **28-1** in response to a rotational force of the object (e.g., the door via half of a hinge object **12-3**). The threaded plunger **79** is further configured to generate the inward force when a nut **81** rotates on threads of the threaded plunger as a result of the rotational force of the object (e.g., closing the door). The threaded plunger is further configured to facilitate an outward force to the plunger **28-1** in response to a counter-rotational force of the object. The threaded plunger is further configured to generate the outward force when the nut counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object (e.g., opening the door).
[0333] The hinge module **204** further includes a fastener **83** to attach the nut **81** to the object **12-3** in a fixed relative position. As such, when the object **12-3** rotates versus object **12-2** (e.g., a portion

of the hinge affixed to a doorjamb) the not anyone rotates around the threaded non-rotational plunger **79** thus moving the plunger laterally.

[0334] The hinge module **204** further includes a second plunger **28-2**. The second plunger is configured to produce the inward force from the threaded plunger to the plunger of the STF module **206** when the nut rotates on threads of the threaded plunger as the result of the rotational force of the object. The second plunger is further configured to produce the outward force from the threaded plunger to the plunger when the nut counter-rotates on the threads of the threaded plunger as the result of the counter-rotational force of the object.

[0335] In an embodiment, the inch module **204** further includes a stop adjuster **87**. The stop adjuster **87** stops the threaded non-rotational plunger **79** from going too far at one end to facilitate a doorstop function.

[0336] The hinge module **204** further includes a fixed bushing **77** to guide the second plunger **28-2** from the threaded plunger to the plunger **28-1** and **61** of the STF module **206**. In an embodiment, the fixed bushing **77** is attached to a rotating chamber **73-2** of the object **12-3** utilizing a fastener **83**. In an embodiment, the object **12-2** further includes a fixed chamber **75** to encircle the fixed bushing **77**. In another embodiment, the fixed chamber **75** is a separate component of the hinge module **204** and is not part of the object **12-2**.

[0337] The plunger **79** is hard coupled to the plunger **28-2** and the plunger **28-2** utilizes a dowel key **85** within a slot of the fixed bushing **77** to prevent the combination of the threaded plunger and second plunger **28-2** from rotating and instead the rotation of the nut **81** around the threaded plunger causes the combination of the threaded plunger and the second plunger to move in a lateral fashion in the inward or outward directions through the fixed bushing **77** to make contact with the cap **61** and the plunger **28-1**.

[0338] In an embodiment, the STF module **206** further includes a set of fluid flow sensors positioned proximal to the chamber as previously discussed. The set of fluid flow sensors is configured to provide a fluid response from the STF. The set of fluid flow sensors includes one or more of a valve opening detector associated with a slot set of the piston, a mechanical position sensor, an image sensor, a light sensor, an audio sensor, a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.

[0339] In an embodiment, the STF module **206** further includes a set of fluid manipulation emitters positioned proximal to the chamber. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF, the set of gates, and a slot set of the piston to provide the control of the motion of the object. The set of fluid manipulation emitters includes one or more of a variable flow valve associated with the slot set, a mechanical vibration generator, an image generator, a light emitter, an audio transducer, a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter.

[0340] The set of gates associated with the piston is further configured to, when the set of gates is traveling through the chamber in an outward direction towards the back channel **24**, a first shear threshold effect includes the first range of shear rates when the STF is configured to have the decreasing viscosity. When the set of gates is traveling through the chamber in an inward direction away from the back channel **24**, a second shear threshold effect includes the second range of shear rates when the STF is configured to have the increasing viscosity.

[0341] The gates **57-1** and **57-2** of the STF module **206** are configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel. The STF module **206** further includes a set of hinges **59-1** and **59-2**. The set of hinges is configured to enable the set of gates to swing to the open position from the closed position and to swing to the closed position from the open position.

[0342] FIG. **13**C further illustrates the embodiment of the STF hinge unit **202** that includes the STF module **206** coupled to the hinge module **204** utilizing a retaining ring **210**. In an example of operation, when the set of hinges **59-1** and **59-2** is configured to enable the set of gates to swing to the open position within the front channel, the STF module **206** enables the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in the open position. The STF module **206** enables the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in the closed position.

[0343] In another example of operation, when the set of hinges **59-1** and **59-2** is configured to enable the set of gates to swing to the open position within the front channel, the piston is further configured to facilitate the flow of the STF from the back channel through the slot set of the piston to the front channel to cause the set of gates to operate in the open position and to cause the STF within the back channel to provide the first range of shear rates when the set of gates is operating in the open position.

[0344] The piston is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the front channel through the slot set of the piston to the back channel to cause the STF within the front channel to provide the second range of shear rates when the set of gates is operating in the closed position (e.g., as shown where rotation of object **2-3** turns the knot which turns the threaded plunger which pushes up on the second plunger which pushes up on the and plunger, which pushes up on the piston and gates into the front channel thus compressing the STF, and with a sufficient shear threshold, provides the second range of shear rates and a higher viscosity which slows down the moving parts including a door attached to object **12-3**.

[0345] FIG. **13**D illustrates another embodiment of a head unit system that includes a shear thickening fluid (STF) module **206-1**, a second STF module **206-2**, and a hinge module **204-1**. The STF module **206-1** is coupled to the hinge module **204-1** utilizing a retaining ring **210-1**. The hinge module **204-1** is coupled to the second STF module **206-2** utilizing a second retaining ring **210-2**.

[0346] The STF module **206-1** includes the STF, where the STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0347] The STF module **206** further includes a chamber, where the chamber configured to contain a portion of the STF. The chamber includes a front channel and a back channel. The STF **206** module further includes a set of gates, where the set of gates is configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel. The set of gates is further configured to enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in an open position. The set of gates is further configured to enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in a closed position.

[0348] The STF module **206-1** further includes a piston (e.g., a cupped piston **55-1**), where the piston housed at least partially radially within the chamber. The piston is configured to include a slot set. The piston is further configured to facilitate flow of the STF through the slot set between the front channel and the back channel when the piston travels longitudinally within the chamber.

[0349] The STF module **206-1** further includes a plunger **28-1**, the plunger housed at least partially radially within the back channel of the chamber. The plunger is configured to exert force on the piston in response to the motion of the object.

[0350] The second STF module **206-2** further includes second shear thickening fluid (STF). The second STF is configured to have a decreasing viscosity in response to a third range of shear rates and an increasing viscosity in response to a fourth range of shear rates. The fourth range of shear rates are greater than the third range of shear rates.

[0351] The second STF module **206-2** further includes a second chamber. The second chamber is configured to contain a portion of the second STF. The second chamber includes a second front channel and a second back channel.

[0352] The second STF module **206-2** further includes a second set of gates. The second set of gates is configured to separate the second front channel and the second back channel to control velocity of flow of the second STF between the second front channel and the second back channel. The second set of gates is further configured to enable the second STF to provide the third range of shear rates as the second STF flows from the second back channel to the second front channel when the second set of gates is in an open position. The second set of gates is further configured to enable the second STF to provide the third range of shear rates as the second STF flows from the second front channel to the second back channel when the second set of gates is in a closed position.

[0353] The second STF module **206-2** further includes second piston (e.g., a cupped piston **55-3**). The second piston housed at least partially radially within the second chamber. The second piston configured to include a second slot set. The piston is further configured to facilitate flow of the second STF through the second slot set between the second front channel and the second back channel when the piston travels longitudinally within the second chamber.

[0354] The second STF module **206-2** further includes a second plunger **28-4**. The second plunger housed at least partially radially within the second back channel of the second chamber. The second plunger is configured to exert force on the second piston in response to the motion of the object.

[0355] The hinge module **204-1** includes a threaded plunger (e.g., a threaded non-rotational plunger **79-1**). The threaded plunger is configured to facilitate an inward force to the plunger **28-1** via a plunger **28-2** of the hinge module **204-1** and a simultaneous outward force to the second plunger **28-4** via a plunger **28-3** in response to a rotational force of the object (e.g., the door). The threaded plunger is further configured to generate the inward force to the plunger and the simultaneous outward force to the second plunger when a nut **81** rotates on threads of the threaded plunger as a result of the rotational force of the object. The nut **81** is coupled to object **12-2** of the hinge module **204-1** utilizing a fastener **83** such that the nut **81** turns on the threaded plunger when a combination of the nut **81** and the object **12-2** rotates (e.g., closing the door).

[0356] The threaded plunger is further configured to facilitate an outward force to the plunger **28-1** via the plunger **28-2** and a simultaneous inward force to the second plunger **28-4** via the plunger **28-3** in response to a counter-rotational force of the object. The threaded plunger is further configured to generate the outward force to the plunger and the simultaneous inward force to the second plunger when the nut **81** counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object (e.g., object **12-2** and the door).

[0357] The hinge module **204-1** further includes object **12-3** (e.g., a portion of the hinge connected to a doorjamb). The object **12-3** is coupled to fixed bushings **77-1** and **77-2** at an upper and lower position of the object **12-3**, where the fixed bushing **77-1** guides the plunger **28-2** and the fixed bushing **77-2** guides the plunger **28-3**. The combination of the plunger **28-3**, the threaded non-rotational plunger **79-1** and the plunger **28-2** is forced to travel in the inward and outward directions without rotation utilizing a dowel key **85-1** within a slot of the plunger **28-2** and another dowel key **85-2** within a slot of the plunger **28-3**.

[0358] The STF of the STF modules **206-1** and **206-2** may have different properties to produce custom results for the overall operation of the head unit system. When the gates and cupped pistons are implemented facing both the outward ends of the head unit system, the STF of the STF module **206-1** may be utilized to provide the second range of shear rates to control the door closing and the STF of the STF module **206-2** may be utilized to provide the second range of shear rates to control the door opening.

[0359] In an alternative embodiment, the cupped piston **55-2** and the gates and hinges associated with the cupped piston **55-2** may be flipped such that both STF modules provide similar motion control when utilizing similar STF. Within the STF module **206-2**, when the set of hinges is configured to enable the set of gates to swing to the open position within the back channel of the STF module **206-2**, the STF provides the second range of shear rates as the STF flows from the

back channel to the front channel when the set of gates is in the closed position and the STF provides the first range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in the open position.
[0360] Further within the STF module **206-2**, when the set of hinges is configured to enable the set of gates to swing to the open position within the back channel, the piston is further configured to facilitate the flow of the STF from the front channel through the slot set of the piston to the back channel to cause the set of gates to operate in the open position and to cause the STF within the front channel to provide the first range of shear rates when the set of gates is operating in the open position. The piston is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the back channel through the slot set of the piston to the front channel to cause the STF within the back channel to provide the second range of shear rates when the set of gates is operating in the closed position.
[0361] FIGS. **14**A-**14**C are schematic block diagrams of another embodiment of a mechanical system (e.g., a head unit system) to control an object (e.g., a door). FIGS. **14**A and **14**B illustrate embodiments where the head unit system includes the STF module **206-1** and the hinge module **204**.
[0362] The STF module **206-1** includes a shear thickening fluid (STF). The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.
[0363] The STF module **206-1** further includes a chamber **16**, where the chamber is configured to contain a portion of the STF. The chamber includes a front channel and a back channel. A backstop **51-1** contains the STF in the front channel.
[0364] The STF module **206-1** further includes a set of gates, where the set of gates is configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel. The set of gates is further configured to enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in an open position. The set of gates is further configured to enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in a closed position.
[0365] The STF module **206-1** further includes a piston, where the piston is housed at least partially radially within the chamber **16**. The piston is configured to include a slot set. The piston is further configured to facilitate flow of the STF through the slot set between the front channel and the back channel when the piston travels longitudinally within the chamber **16**.
[0366] The STF module **206-1** further includes a plunger **28-1**. The plunger is housed at least partially radially within the back channel of the chamber. The plunger is configured to exert force on the piston in response to the motion of the object. The STF module **206-1** further includes a cap **61** to operate in conjunction with the plunger **28-1** to exert pressure on the plunger. A spring **45** stores energy and releases the energy by pushing the cap **61** outward from the chamber **16** to pull the piston via the plunger from the front channel to the back channel.
[0367] The STF module **206-1** further includes a modular castle mount **220**. The modular castle mount is configured to extend the chamber to facilitate the configuration of the plunger to exert the force on the piston in response to the motion of the object. The modular castle mount **220** includes a set of teeth and/or tabs constructed in a circle around the diameter of the chamber **16**. Each of the set of teeth and/or tabs includes a retaining ring channel **212**.
[0368] The hinge module **204** includes a threaded plunger, where the threaded plunger is configured to facilitate an inward force to the plunger in response to a rotational force of the object. The threaded plunger is further configured to generate the inward force when a nut rotates on threads of the threaded plunger as a result of the rotational force of the object. The threaded plunger is further configured to facilitate an outward force to the plunger in response to a counter-rotational

force of the object. The threaded plunger is further configured to generate the outward force when the nut counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object.

[0369] The hinge module **204** further includes a reciprocal modular castle mount **220**, where the reciprocal modular castle mount is configured to couple the reciprocal modular castle mount with the modular castle mount. The threaded plunger is configured to apply the inward force to the plunger in response to the rotational force of the object when the reciprocal modular castle mount is coupled with the modular castle mount. The threaded plunger is further configured to apply the outward force to the plunger in response to the counter-rotational force of the object when the reciprocal modular castle mount is coupled with the modular castle mount.

[0370] The reciprocal modular castle mount includes a reciprocal set of teeth and/or reciprocal tabs constructed in a circle around the diameter of a portion of the hinge module **204** (e.g., a rotating chamber **73-1** or a fixed chamber **75**). Each of the set of teeth and/or tabs includes a retaining ring channel **212** such that a single common retaining ring channel is formed when the reciprocal modular castle mount is coupled with the modular castle mount. A retaining ring **210** configured within the single common retaining ring channel facilitates coupling of the STF module **206-1** and the hinge module **204** as illustrated in FIG. **14**B.

[0371] The hinge module **204** further includes a lower castle mount **221** to facilitate coupling of the STF module **206-2** of FIG. **13**D with the hinge module **204** utilizing another retaining ring **210-2** as illustrated in FIG. **14**C. The STF module **206-2** further includes a backstop **51-2** to contain the STF of the STF module **206-2** within a front channel of a chamber **16-2**.

[0372] The hinge module **204** further includes one or more fastener holes **222** to accept one or more fasteners **226** of FIG. **14**B to secure the object **12-3** and to secure the object **12-2**. The objects **12-3** and **12-2** further include one or more mounting holes **224** to secure the objects **12-3** and **12-2** to other objects.

[0373] FIGS. **15**A-**15**C are schematic block diagrams of another embodiment of a mechanical system (e.g., a head unit system) to control an object. The head unit system includes the hinge module **204** and STF module **206-1** of FIG. **14**A and one of accessory modules **208-1** through **208-7**. The accessory modules **208-1** through **208-7** also couple to the hinge module **204** utilizing the castle mount **220** as discussed in reference to FIGS. **14**A-C.

[0374] Each accessory module includes an accessory coupling mechanism that is configured to couple the accessory module to the hinge module. For example, the accessory coupling mechanism is configured with a second modular castle mount that couples the accessory module to the hinge module via a second reciprocal modular castle mount of the hinge.

[0375] Each accessory module further includes a computing device configured to operate in accordance with a first mode of the first range of shear rates. The computing device is further configured to operate in accordance with a second mode of the second range of shear rates when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism.

[0376] The computing device is configured with a processor for executing processor-executable instructions and a storage device for storing the processor-executable instructions. The processor-executable instructions include a first set of processor-executable instructions that are configured to facilitate the first mode of the first range of shear rates and a second set of processor-executable instructions that are configured to facilitate the second mode of the second range of shear rates.

[0377] The first set of processor-executable instructions is further configured to facilitate the first mode of the first range of shear rates, when executed by the processor, to cause the processor to perform one or more of a variety of functions. A first function includes controlling a motor configured to accelerate the rotational force of the object. For example, the accessory module includes the motor and is activated when the object (e.g., a door) is moving slowly (e.g., the first range) to speed up closing the door.

[0378] A second function includes determining a position of the object. For example, the processor calculates the position of the door from sensor data when the accessory module includes a position sensor. A third function includes determining a movement vector of the object. For example, the processor calculates movement vector of the door based on time and the position of the door from the sensor data when the accessory module includes the position sensor.
[0379] A fourth function includes identifying a secondary object associated with the object. For example, the processor identifies a person or a pet (e.g., near the door) from an image sensor output and subsequently activates the motor to contain the pet to a particular area. A fifth function includes interpreting a request to change the movement vector of the object. For example, the processor decodes an IoT command via a wireless link to stop the door or even close the door. A sixth function includes indicating a status of the object. The status of the object includes one or more of open (e.g., door open), closed, locked, unlocked, safe (e.g., when no one is blocking the door, or no fire on the other side of the door, etc.), not safe (e.g., person in way, pet in way, intruder in next room, active shooter in next room or nearby, fire or other hazard present), and emergency fast closing when the object is a door.
[0380] In another embodiment, a seventh function includes controlling a set of fluid manipulation emitters positioned proximal to the chamber to facilitate the first range of shear rates. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF, the set of gates, and a slot set of the piston to provide the control of the motion of the object (e.g., to quickly close the door).
[0381] The second set of processor-executable instructions is further configured to facilitate the second mode of the second range of shear rates (e.g., higher viscosity of the STF), when executed by the processor, to cause the processor to perform one or more of a variety of other functions. A first function includes controlling the motor configured to deaccelerate the rotational force of the object (e.g., stop the door for safety reasons or slow close to allow someone to enter within a short timeframe). A second function includes detecting an end-position of the object (e.g., the door has closed). A third function includes determining an end-movement vector of the object (e.g., based on speed, position, and time of just finishing closing the door).
[0382] A fourth function includes determining a movement vector of the secondary object associated with the object (e.g., estimating speed and position of the pet or person based on an image sensor output). A fifth function includes interpreting a request to change the end-movement vector of the object (e.g., the processor decodes another IoT command via the wireless link to slow down the closing of the door even more or keep it just open).
[0383] A sixth function includes indicating a further status of the object. The further status of the object includes one or more of emergency slow closing and locking, when the object is the door. For instance, in response to a wireless message from a device initiated by a teacher, the processor activates the motor to quickly close a classroom door and indicate the emergency closing status. In an embodiment, a seventh function includes controlling the set of fluid manipulation emitters to facilitate the second range of shear rates (e.g., to slow close the door).
[0384] A technological improvement is provided to the STF module **206-1** with regards to controlling motion of the object by utilizing one of the accessory modules **208-1** through **208-7**. The STF module **206-1** and accessory module share the hinge module **204** and operate within proximity. In an embodiment, each of the STF module **206-1** and the accessory modules **208-1** through **208-7** include the computing device **100-1** of FIG. **3** such that the STF module **206-1** communicates with the accessory module via wireless communication to facilitate optimization of the controlling of the motion of the object.
[0385] The accessory modules **208-1** through **208-7** represent a few examples of numerous embodiments of the accessory module to provide one or more specialty sensors and/or actuators with regards to the operation of the head unit system. For example, the accessory module **208-1** includes a media sensor such as one or more of an image sensor and a microphone.

[0386] As another example, the accessory module **208-2** includes a set of wireless communication modems to provide communications between modules sharing the hinge module **204** depicted in FIG. **14**C and as discussed with reference to FIG. **15**B and with other modules associated with other hinge modules as discussed further with reference to FIG. **15**C. For instance, an infrared wireless communication path is established through the hinge module **204** between the STF module **206-1** mounted on the castle mount **220** and the accessory module **208-2** mounted on another castle mount **220** of the hinge module **204**. In another instance, the infrared wireless communication path is further established to module associated with another hinge module **204**. In yet another instance, a Wi-Fi wireless communication path is established from the accessory module **208-2** to another module in another room.
[0387] As another example, the accessory module **208-3** includes visual user output for status and other purposes via a set of status light emitting diodes (LED). For example, illumination of the LED with a red color indicates that an associated door is open while elimination of the LED with a green color indicates that the associated door is closed.
[0388] As another example, the accessory module **208-4** includes an energy harvesting apparatus to produce minute amounts of energy that is temporarily stored (e.g., in a battery, in a capacitor, etc.) as a result of motion of the object. The energy may be utilized to power the computing device **100-1** of the same module. In another embodiment, the energy may be utilized power another module coupled to the same hinge module **204**, where an isolated center conductor that runs through the hinge module **204** transfers the energy from the energy affiliated accessory module **208-4** to the other module.
[0389] As yet another example, the accessory module **208-5** includes an Internet of Things (IoT) functionality to both sense operations of the STF module **206-1** and control the STF module **206-1**. In an instance, the IoT functionality includes providing status of a door associated with the head unit system to a digital twin platform. In another instance, the IoT functionality includes providing a security service for the door associated with the head unit system (e.g., triggering an alarm for an unexpected opening of the door, indicating when the second range of shear rates are utilized within the STF module **206-1**, etc.).
[0390] As a further example, the accessory module **208-6** includes an electric motor such that the electric motor can affect the position and motion of the object. For instance, the electric motor is activated to further slow down the closing of the door in addition to a safety response from the STF module **206-1**. In another instance, the electric motor is activated to open a closed door to assistant individual that normally needs assistance to open doors. In yet another instance, the motor stops the travel of all the mechanicals when the motor is implemented as a solenoid (e.g., a coil around a solenoid plunger) where the solenoid plunger travels into any rotating portion of the head unit system to immediately stop rotation stopping movement of the object (e.g., holding a closed door closed/locked, preventing a door from opening more than a desired maximum amount, etc.).
[0391] As a still further example, the accessory module **208-7** includes a position sensor to provide information with regards to the position and/or motion of the object. For instance, the position sensor indicates how far open the door for comparison to a security threshold level of opening of the door. In another instance, the position sensor is utilized to track exactly how the head unit system operates when the second range of shear rates are active (e.g., how effective is the STF module **206-1** at providing a desired safe and quiet closing of the door).
[0392] FIG. **15**B illustrates scenario examples of utilization of the STF module **206-1**, the hinge module **204**, and one of a variety of accessory modules that shares the hinge module **204** in common with the STF module. The head unit system for controlling motion of an object includes a shear thickening fluid (STF) module, a hinge module, and an accessory module.
[0393] The STF module includes a shear thickening fluid (STF), where the STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first

range of shear rates. The STF module further includes a chamber, where the chamber is configured to contain a portion of the STF. The chamber includes a front channel and a back channel as previously discussed. The STF module further includes a set of gates, where the set of gates is configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel. The set of gates is further configured to enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in an open position, wherein the set of gates is further configured to enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in a closed position.

[0394] The STF module further includes a piston, where the piston is housed at least partially radially within the chamber. The piston is further configured to include a slot set. The piston is further configured to facilitate flow of the STF through the slot set between the front channel and the back channel when the piston travels longitudinally within the chamber.

[0395] The STF module further includes a plunger, where the plunger is housed at least partially radially within the back channel of the chamber. The plunger is configured to exert force on the piston in response to the motion of the object.

[0396] The hinge module **204** includes a coupling mechanism, where the coupling mechanism is configured to couple the STF module to the hinge module. The hinge module further includes a threaded plunger, where the threaded plunger is configured to facilitate an inward force to the plunger in response to a rotational force of the object. The threaded plunger is further configured to generate the inward force when a nut rotates on threads of the threaded plunger as a result of the rotational force of the object. The threaded plunger is further configured to facilitate an outward force to the plunger in response to a counter-rotational force of the object. The threaded plunger is further configured to generate the outward force when the nut counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object.

[0397] The accessory module includes an accessory coupling mechanism, where the accessory coupling mechanism is configured to couple the accessory module to the hinge module. The accessory module further includes a computing device, where the computing device configured to operate in accordance with a first mode of the first range of shear rates when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism. For example, the computing device monitors the operation of the head unit system when the first mode is to monitor operations while the STF is operating in the first range of shear rates.

[0398] In an embodiment, the accessory module further includes a motor, where the motor is coupled to the threaded plunger and configured to exert a force on the threaded plunger (e.g., to open or close the hinge module and hence open or close a door when a door is the object). The computing device is configured to operate in accordance with another first mode of the first range of shear rates to control the motor to accelerate the rotational force of the object (e.g., to quickly close the door).

[0399] The computing device is further configured to operate in accordance with a second mode of the second range of shear rates when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism. For example, the computing device actively controls parameters of the STF module when the STF is operating in the second range of shear rates.

[0400] In an embodiment, the computing device is further configured to operate in accordance with another second mode of the second range of shear rates to control the motor to deaccelerate the rotational force of the object (e.g., slow close the door) when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism.

[0401] A first example scenario of utilization of the head unit system, that includes the status LED accessory module and the STF module, includes the status LED indicating one or more of open/closed door, locked/unlocked lock, safe/not safe condition, and facilitating monitoring and control of every door closing utilizing each available STF threshold zone. To accommodate multiple status scenarios parameters of the LEDs are modified in correlation with a particular status. For example, a green LED illumination indicates that the door is closed while a red LED illumination indicates that the door is open. A flashing yellow LED indicates that an unsafe condition exists while a blue LED indicates that the doorway is safe. Flash patterns may also be utilized in addition to an illumination level of the LED. For example, a fast flash pattern indicates that the second STF threshold zone has been utilized while a slow flash pattern indicates that only the first STF threshold zone has been utilized.

[0402] A second example scenario of utilization of the head unit system includes the IoT module and the STF module. The IoT logic can be a wireless communication modem powered by an internal door movement energy harvesting device provides IoT logic and communications of door status with other similar devices and determines door control functions for storage and execution. For example, the IoT module is preprogrammed with parameters of the STF of the STF module such that when monitoring position and velocity of door movement the IoT module can predict which zone the STF is operating in to report STF utilization information to another computing device (e.g., of a facility where the head unit system is utilized along with numerous other head unit systems for controlling doors.

[0403] A third example scenario of utilization of the head unit system includes the motor module and the STF module. An electric motor functions to rotate one of the objects associated with the hinge objects to affect repositioning of an associated door. Movement via the electric motor is coordinated with the parameters of the STF module to provide a composite desire door functionality with regards to speed of closing the door and overall safety functions. In an instance, the computing device associated with the motor module interprets the movement of the door over numerous door opening and closing cycles to estimate the parameters of the STF (e.g., shear threshold ranges and zones versus viscosity). Once the parameters are determined, the computing device of the motor module determines when and how to activate the electric motor to assist the STF module to provide an overall desired opening and closing pattern for safety and convenience purposes.

[0404] A fourth example scenario of utilization of the head unit system includes the position sensor and the STF module. For example, a precision linear potentiometer is utilized within the position sensor module such that a computing device associated with the precision linear potentiometer interprets an output of the precision linear potentiometer to produce position, velocity, and acceleration information with regards to all physical aspects of the hinge module **204**. In an instance, computing device of the position sensor monitors movement of the hinge module over numerous opening and closing cycles to compare the response of the STF module for each of the opening and closings. The comparisons are interpreted to determine whether the STF module is still operating within desired operational parameters or if the STF module needs an adjustment or replacement. When a threshold level of undesired performance is reached, the position sensor module provides an indication to the head unit system and/or to at least one other head unit system for potential corrective action.

[0405] In another instance, the computing device of the position sensor module further monitors the movement of the hinge module and compares a position and velocity signature of such openings and closings to other correlated information (e.g., day of week, time of day, traffic patterns, individual people coming through the doorway, etc.) to provide an output that is an estimate of an outcome associated with the other correlated information. For example, the position sensor module indicates that Joe has just walked through the door based on patterns of utilization of the door by Joe over the last **200** utilizations by Joe.

[0406] FIG. **15**C illustrates scenario examples of utilization of at least two of the STF module **206-**

1, the hinge module **204**, and one of a variety of accessory modules that shares the hinge module **204**. For example, the head unit system for controlling motion of an object includes a first STF module, a first hinge module coupled to the object and the first STF module, a second hinge module also coupled to the object, and an accessory module coupled to the second hinge module, where the accessory module is configured to interact with the first STF module (e.g., communicate, share power sources, control the object in a coordinated way, etc.)

[0407] A first scenario example of the utilization of two or more hinge modules along with multiple other modules includes the status LED module, the IoT module, the media sensor module, and the STF module. For example, the IoT module interprets image sensor imagery to provide improved status for display by the status LED and control of the door via adapting the STF. For instance, the image sensor mounted on the lower hinge constantly views foot traffic associated with people utilizing the door to produce statistics (e.g., traffic pattern versus time of day, velocity of entering the doorway, velocity of exiting the doorway, force utilized by each person to operate the door in either direction, etc.) associated with the aggregate of people utilizing the door. The IoT module determines implications of the traffic patterns on safety factors associated with utilizing the door and issues status information to the status LED module for display. For instance, the status LED module flashes the yellow LED with a fast pattern to indicate that parameters of the STF module may not be able to safely meet the needs of an eminent door opening sequence.

[0408] A second scenario example of the utilization of the two or more hinge modules along with multiple other modules includes the motor module, the position sensor module, the media sensor module, and the STF module. For example, electric motor module interprets output from the position sensor module and output from the image sensor of the media sensor module to provide improved adaptation of the STF of the STF module. For instance, the computing device of the motor module interprets the position sensor output over time and compares a desired response of the system versus actual to identify gaps in operation where the electric motor can assist by applying an opening or a closing force to the door.

[0409] A third scenario example of the utilization of the two or more hinge modules along with multiple other modules includes the media sensor module, the IoT module, the motor module, and the STF module, where the media sensor module is mounted at an upper hinge location. For example, an image sensor associated with the upper mounted media sensor module is utilized to perform routine facial recognition of people utilizing the doorway for correlation with typical parameters associated with operation of the door (e.g., force of opening, force of closing, speed of door operation, etc.). The IoT module acquires further door utilization information from another head unit system associated with another door for a common person that utilizes both doorways. The computing device of the electric motor module summarizes the door utilization information for the common person to produce adaptation commands for the electric motor to assist the STF module in the operation of the door to further benefit the common person as they utilize this particular door. For instance, recognizing that Joe is approaching the door utilizing the media sensor, the motor module applies energy to the electric motor to assist in pre-positioning the door in conjunction with the STF module to provide a better experience for Joe and utilizing the door as Joe desires.

[0410] In another instance of the third scenario example, the IoT module interprets an emergency lock signal from another wireless device (e.g., a wireless pendent utilized by a school teacher). When the emergency lock signal is detected, the motor module operates a lockdown of the door movement (e.g., closed a hold, hold).

[0411] In another scenario, the media sensor output is interpreted by the computing device of the media sensor to identify an unsafe local condition (e.g., fire, unauthorized access, water on the floor, someone standing in the way of the way the door swings on the other side of the door, etc.). The IoT module communicates the unsafe local condition to at least one other head unit system associated with another door for subsequent optimization. The electric motor of the motor module prevents the door from opening due to the unsafe local condition.

[0412] In yet another scenario, the computing device associated with any of the modules (e.g., the STF module) interprets fluid response from a set of fluid flow sensors associated with the chamber to produce a piston velocity and a piston position of the piston associated with the STF module. The computing device determines a shear force based on the piston velocity and the piston position. The computing device determines a desired response of the STF based on one or more of the shear force, the piston velocity, and the piston position. The computing device activates an accessory function in accordance with the desired response for the STF to cause selection of one of the first range of shear rates and the second range of shear rates. Examples of the accessory function include a status LED, and IoT function, a motor control function, a position determination function, an image processing function, a sound processing function, etc.

[0413] FIG. **15**D illustrates a further scenario example of utilization of at least two of the STF module **206-1**, the hinge module **204**, and one of a variety of accessory modules that shares the hinge module **204** on three doors **302-1** through **302-3** within a hallway **300**. For example, the three head unit systems each include an STF module, a status LED module, a motor module, and a media sensor module. The scenario includes an active shooter situation where the shooter is closer to doors **302-1** and **302-2** and further away from door **302-3**. The scenario further includes a person near door **302-3**.

[0414] The media sensor module includes a detection function to detect the active shooter scenario. The detection function includes at least one of a gunshot detector (e.g., matching live audio samples to known gunshot audio patterns while excluding routine sounds) and an image matcher to identify a weapon and/or the shooter holding the weapon (e.g., matching live images/video samples to stored images/video of weapons and shooters with weapons).

[0415] The processor associated with the media sensor of door **302-1** detects that a safety threat exists within a first proximity threshold of the object. The safety threat includes anything that threatens the safety of people or property. Examples include an active shooter, a fire, a building collapse, and a person intent on causing bodily harm. The detecting includes local detection by sensing and receiving a message. For example, a gunshot is detected by the processor associated with the media sensor of door **302-1** and receives a wireless message from the processor associated with the media sensor of door **302-3** that the gunshot was detected slightly later.

[0416] The processor associated with the motor of the door **302-1** controls the motor configured to accelerate the rotational force of the object when the object **302-1** is the door while the door closes with a first velocity (e.g., very fast relative to normal operation). The processor associated with the status LED module of door **302-1** indicates a status of the object to include closing of the door with the first velocity (e.g., flashes red to telegraph the urgency of closing the door very fast relative to normal operation). The modules associated with the door **302-2** operate in a similar way as described with reference to the modules associated with the door **302-1**.

[0417] The processor associated with the media sensor of door **302-3** detects that the safety threat exists within a second proximity threshold of the object (e.g., third door). The second proximity threshold is a greater distance than the first proximity threshold. For example, the gunshot is detected by the processor associated with the media sensor of door **302-3** and receives the wireless message from the processor associated with the media sensor of door **302-1** that the gunshot was detected slightly sooner at door **1** as compared to at door **3**.

[0418] The processor associated with the motor of the door **302-3** controls the motor configured to deaccelerate the rotational force of the object when the object **302-1** is the door while the door closes with a second velocity (e.g., similar to normal operation). The processor associated with the status LED module of door **302-3** indicates a status of the object to include closing of the door with the second velocity (e.g., flashes yellow to telegraph the closing the door due to the safety threat).

[0419] In another example of operation, the media sensor of the door **302-3** detects that a person, not the shooter, is nearby the third door and that a safety improvement can be provided by enabling that person to pass through the third door to better relative safety. Having detected the person, the processor of the motor of the door **302-3** controls the motor to accommodate getting that person through the door and then quickly closing the door to protect those behind door **3** from the active shooter. Once the door is closed, the processor of the status LED causes the LED to indicate the door locked (e.g., steady red).

[0420] It is noted that terminologies as may be used herein such as bit stream, stream, signal sequence, etc. (or their equivalents) have been used interchangeably to describe digital information whose content corresponds to any of a number of desired types (e.g., data, video, speech, text, graphics, audio, etc. any of which may generally be referred to as ‘data’).

[0421] As may be used herein, the terms “substantially” and “approximately” provides an industry-accepted tolerance for its corresponding term and/or relativity between items. For some industries, an industry-accepted tolerance is less than one percent and, for other industries, the industry-accepted tolerance is 10 percent or more. Other examples of industry-accepted tolerance range from less than one percent to fifty percent. Industry-accepted tolerances correspond to, but are not limited to, component values, integrated circuit process variations, temperature variations, rise and fall times, thermal noise, dimensions, signaling errors, dropped packets, temperatures, pressures, material compositions, and/or performance metrics. Within an industry, tolerance variances of accepted tolerances may be more or less than a percentage level (e.g., dimension tolerance of less than +/−1%). Some relativity between items may range from a difference of less than a percentage level to a few percent. Other relativity between items may range from a difference of a few percent to magnitude of differences.

[0422] As may also be used herein, the term(s) “configured to”, “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as “coupled to”.

[0423] As may even further be used herein, the term “configured to”, “operable to”, “coupled to”, or “operably coupled to” indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term “associated with”, includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

[0424] As may be used herein, the term “compares favorably”, indicates that a comparison between two or more items, signals, etc., provides a desired relationship. For example, when the desired relationship is that signal **1** has a greater magnitude than signal **2**, a favorable comparison may be achieved when the magnitude of signal **1** is greater than that of signal **2** or when the magnitude of signal **2** is less than that of signal **1**. As may be used herein, the term “compares unfavorably”, indicates that a comparison between two or more items, signals, etc., fails to provide the desired relationship.

[0425] As may be used herein, one or more claims may include, in a specific form of this generic form, the phrase “at least one of a, b, and c” or of this generic form “at least one of a, b, or c”, with more or less elements than “a”, “b”, and “c”. In either phrasing, the phrases are to be interpreted identically. In particular, “at least one of a, b, and c” is equivalent to “at least one of a, b, or c” and shall mean a, b, and/or c. As an example, it means: “a” only, “b” only, “c” only, “a” and “b”, “a” and “c”, “b” and “c”, and/or “a”, “b”, and “c”.

[0426] As may also be used herein, the terms “processing module”, “processing circuit”, “processor”, “processing circuitry”, and/or “processing unit” may be a single processing device or

a plurality of processing devices. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. The processing module, module, processing circuit, processing circuitry, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, processing circuitry, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, processing circuitry, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network). Further note that if the processing module, module, processing circuit, processing circuitry and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, processing circuitry and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.
[0427] One or more embodiments have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims. Further, the boundaries of these functional building blocks have been arbitrarily defined for convenience of description. Alternate boundaries could be defined as long as the certain significant functions are appropriately performed. Similarly, flow diagram blocks may also have been arbitrarily defined herein to illustrate certain significant functionality.
[0428] To the extent used, the flow diagram block boundaries and sequence could have been defined otherwise and still perform the certain significant functionality. Such alternate definitions of both functional building blocks and flow diagram blocks and sequences are thus within the scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules, and components herein, can be implemented as illustrated or by discrete components, application specific integrated circuits, processors executing appropriate software and the like or any combination thereof.
[0429] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with one or more other routines. In addition, a flow diagram may include an “end” and/or “continue” indication. The “end” and/or “continue” indications reflect that the steps presented can end as described and shown or optionally be incorporated in or otherwise used in conjunction with one or more other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram

indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0430] The one or more embodiments are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical embodiment of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the embodiments discussed herein. Further, from figure to figure, the embodiments may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.

[0431] Unless specifically stated to the contra, signals to, from, and/or between elements in a figure of any of the figures presented herein may be analog or digital, continuous time or discrete time, and single-ended or differential. For instance, if a signal path is shown as a single-ended path, it also represents a differential signal path. Similarly, if a signal path is shown as a differential path, it also represents a single-ended signal path. While one or more particular architectures are described herein, other architectures can likewise be implemented that use one or more data buses not expressly shown, direct connectivity between elements, and/or indirect coupling between other elements as recognized by one of average skill in the art.

[0432] The term “module” is used in the description of one or more of the embodiments. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more sub-modules, each of which may be one or more modules.

[0433] As may further be used herein, a computer readable memory includes one or more memory elements. A memory element may be a separate memory device, multiple memory devices, or a set of memory locations within a memory device. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, a quantum register or other quantum memory and/or any other device that stores data in a non-transitory manner. Furthermore, the memory device may be in a form of a solid-state memory, a hard drive memory or other disk storage, cloud memory, thumb drive, server memory, computing device memory, and/or other non-transitory medium for storing data. The storage of data includes temporary storage (i.e., data is lost when power is removed from the memory element) and/or persistent storage (i.e., data is retained when power is removed from the memory element). As used herein, a transitory medium shall mean one or more of: (a) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for temporary storage or persistent storage; (b) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for temporary storage or persistent storage; (c) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for processing the data by the other computing device; and (d) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for processing the data by the other element of the computing device. As may be used herein, a non-transitory computer readable memory is substantially equivalent to a computer readable memory. A non-transitory computer readable memory can also be referred to as a non-transitory computer readable storage medium.

[0434] While particular combinations of various functions and features of the one or more embodiments have been expressly described herein, other combinations of these features and functions are likewise possible. The present disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

Claims

1. A head unit system for controlling motion of an object, the head unit system comprising: a shear thickening fluid (STF) module, wherein the STF module includes: a shear thickening fluid (STF), wherein the STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates, wherein the second range of shear rates are greater than the first range of shear rates; a chamber, the chamber configured to contain a portion of the STF, wherein the chamber includes a front channel and a back channel; a piston, the piston housed at least partially radially within the chamber, the piston configured to include a slot set, wherein the piston is further configured to facilitate flow of the STF through the slot set between the front channel and the back channel when the piston travels longitudinally within the chamber; and a plunger, the plunger housed at least partially radially within the back channel of the chamber, wherein the plunger is configured to exert force on the piston in response to the motion of the object; a hinge module, wherein the hinge module includes: a coupling mechanism, the coupling mechanism configured to couple the STF module to the hinge module; and a threaded plunger, the threaded plunger configured to facilitate an inward force to the plunger in response to a rotational force of the object, wherein the threaded plunger is further configured to generate the inward force when a nut rotates on threads of the threaded plunger as a result of the rotational force of the object, wherein the threaded plunger is further configured to facilitate an outward force to the plunger in response to a counter-rotational force of the object, wherein the threaded plunger is further configured to generate the outward force when the nut counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object; and an accessory module, wherein the accessory module includes: an accessory coupling mechanism, the accessory coupling mechanism configured to couple the accessory module to the hinge module; a motor, the motor coupled to the threaded plunger and configured to exert a force on the threaded plunger; and a computing device, the computing device includes a wireless communication modem and is configured to operate in accordance with a first mode of the first range of shear rates to control the motor to accelerate the rotational force of the object in response to a wireless message received via the wireless communication modem from another computing device, wherein the computing device is further configured to operate in accordance with a second mode of the second range of shear rates to control the motor to deaccelerate the rotational force of the object in response to the wireless message when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism.
2. The head unit system of claim 1, wherein the STF module further comprises: a modular castle mount, the modular castle mount configured to extend the chamber to facilitate the configuration of the plunger to exert the force on the piston in response to the motion of the object.
3. The head unit system of claim 2, wherein the hinge module further comprises: a reciprocal modular castle mount, the reciprocal modular castle mount configured to couple the reciprocal modular castle mount with the modular castle mount, wherein the threaded plunger is configured to apply the inward force to the plunger in response to the rotational force of the object when the reciprocal modular castle mount is coupled with the modular castle mount, wherein the threaded plunger is further configured to apply the outward force to the plunger in response to the counter-rotational force of the object when the reciprocal modular castle mount is coupled with the modular castle mount.
4. The head unit system of claim 3, wherein the accessory module further comprises: the accessory coupling mechanism configured with a second modular castle mount, wherein the second modular

castle mount couples the accessory module to the hinge module via a second reciprocal modular castle mount of the hinge module.

5. The head unit system of claim 1, wherein the slot set further comprises one or more of: a first slot of the slot set configured with at least one cylindrical tube with substantially consistent diameter from one side of the piston to an opposite side of the piston; a second slot of the slot set configured with at least one conical shaped tube with an increasing diameter from the one side of the piston to the opposite side of the piston; a third slot of the slot set configured with the at least one conical shaped tube with a decreasing diameter from the one side of the piston to the opposite side of the piston; and a fourth slot of the slot set configured with at least one venturi shaped tube from the one side of the piston to the opposite side of the piston.

6. The head unit system of claim 1, wherein the STF module further comprises: a set of gates, the set of gates configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel, wherein the set of gates is further configured to enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in an open position, wherein the set of gates is further configured to enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in a closed position; the set of gates is further configured to: when the set of gates is traveling through the chamber in an outward direction towards the back channel, a first shear threshold effect includes: the first range of shear rates when the STF is configured to have the decreasing viscosity, and when the set of gates is traveling through the chamber in an inward direction away from the back channel, a second shear threshold effect includes: the second range of shear rates when the STF is configured to have the increasing viscosity; and a set of hinges, the set of hinges configured to enable the set of gates to swing to the open position from the closed position and to swing to the closed position from the open position; and wherein the set of gates is further configured to: when the set of hinges is configured to enable the set of gates to swing to the open position within the front channel: enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in the open position, and enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in the closed position; and when the set of hinges is configured to enable the set of gates to swing to the open position within the back channel: enable the STF to provide the second range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in the closed position, and enable the STF to provide the first range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in the open position.

7. The head unit system of claim 6, wherein the STF module further comprises: when the set of hinges is configured to enable the set of gates to swing to the open position within the front channel: the piston is further configured to facilitate the flow of the STF from the back channel through the slot set of the piston to the front channel to cause the set of gates to operate in the open position and to cause the STF within the back channel to provide the first range of shear rates when the set of gates is operating in the open position, wherein the piston is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the front channel through the slot set of the piston to the back channel to cause the STF within the front channel to provide the second range of shear rates when the set of gates is operating in the closed position; and when the set of hinges is configured to enable the set of gates to swing to the open position within the back channel: the piston is further configured to facilitate the flow of the STF from the front channel through the slot set of the piston to the back channel to cause the set of gates to operate in the open position and to cause the STF within the front channel to provide the first range of shear rates when the set of gates is operating in the open position, wherein the piston is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the back channel through the slot set of the piston to the front channel to cause the

STF within the back channel to provide the second range of shear rates when the set of gates is operating in the closed position.
8. The head unit system of claim 1, wherein the STF module further comprises: a bushing, the bushing configured in a fixed position relative to the chamber to cause containment of the STF in the back channel; and a backstop, the backstop configured with the chamber to cause containment of the STF in the front channel.
9. The head unit system of claim 1, wherein the STF module further comprises: a cap, the cap configured to move in unison with the plunger radially within the chamber; and a spring, the spring configured to store energy when the cap moves in an inward direction in response to a force from the object, wherein the spring is further configured to release the energy, absent the force from the object, such that the cap and the plunger move in an outward direction.
10. The head unit system of claim 1, wherein the hinge module further comprises: a second plunger, the second plunger configured to produce the inward force from the threaded plunger to the plunger when the nut rotates on threads of the threaded plunger as the result of the rotational force of the object, wherein the second plunger is further configured to produce the outward force from the threaded plunger to the plunger when the nut counter-rotates on the threads of the threaded plunger as the result of the counter-rotational force of the object.
11. The head unit system of claim 1, wherein the STF comprises: a plurality of nanoparticles, wherein the plurality of nanoparticles includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO_2, polystyrene, polymethylmethacrylate, or a mixture thereof.
12. The head unit system of claim 1, wherein the STF further comprises: one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture thereof.
13. The head unit system of claim 1, wherein the STF module further comprises: the computing device configured with: a processor for executing processor-executable instructions, the wireless communication modem configured to communicate a set of wireless messages with a set of other computing devices, an image sensor configured to provide imagery, and a storage device for storing the processor-executable instructions, wherein the processor-executable instructions include: a first set of processor-executable instructions, the first set of processor-executable instructions configured to communicate a first wireless message with regards to first imagery, wherein the set of wireless messages includes the first wireless message, a second set of processor-executable instructions, the second set of processor-executable instructions configured to communicate a second wireless message to control the motor, wherein the set of wireless messages includes the second wireless message, and a third set of processor-executable instructions, the third set of processor-executable instructions configured to communicate a third wireless message to control another motor associated with another accessory module, wherein the set of wireless messages includes the third wireless message.
14. The head unit system of claim 1, wherein the STF module further comprises: a set of fluid flow sensors positioned proximal to the chamber, wherein the set of fluid flow sensors are configured to provide a fluid response from the STF, wherein the set of fluid flow sensors includes one or more of: a valve opening detector associated with a slot set of the piston, a mechanical position sensor, an image sensor, a light sensor, an audio sensor, a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.
15. The head unit system of claim 1, wherein the STF module further comprises: a set of fluid manipulation emitters positioned proximal to the chamber, wherein the set of fluid manipulation emitters provide a fluid activation to at least one of the STF, a set of gates, and a slot set of the piston to provide the control of the motion of the object, wherein the set of fluid manipulation emitters includes one or more of: a variable flow valve associated with the slot set, a mechanical vibration generator, an image generator, a light emitter, an audio transducer, a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency

wireless field transmitter.

16. The head unit system of claim 1, wherein the accessory module further comprises: the computing device configured with: a processor for executing processor-executable instructions, the wireless communication modem configured to communicate the wireless message with the other computing device, and a storage device for storing the processor-executable instructions, wherein the processor-executable instructions include: a first set of processor-executable instructions, the first set of processor-executable instructions configured to facilitate the first mode of the first range of shear rates, and a second set of processor-executable instructions, the second set of processor-executable instructions configured to facilitate the second mode of the second range of shear rates.

17. The head unit system of claim 16, wherein the accessory module further comprises: the first set of processor-executable instructions is further configured to facilitate the first mode of the first range of shear rates, when executed by the processor, to cause the processor to perform one or more of: controlling the motor configured to accelerate the rotational force of the object, determining a position of the object, determining a movement vector of the object, identifying a secondary object associated with the object, interpreting a request to change the movement vector of the object, indicating a status of the object, wherein the status of the object includes one or more of open, closed, locked, unlocked, safe, not safe, and emergency fast closing when the object is a door, and controlling a set of fluid manipulation emitters positioned proximal to the chamber to facilitate the first range of shear rates, wherein the set of fluid manipulation emitters provide a fluid activation to at least one of the STF, the set of gates, and a slot set of the piston to provide the control of the motion of the object; and the second set of processor-executable instructions is further configured to facilitate the second mode of the second range of shear rates, when executed by the processor, to cause the processor to perform one or more of: controlling the motor configured to deaccelerate the rotational force of the object, detecting an end-position of the object, determining an end-movement vector of the object, determining a movement vector of the secondary object associated with the object, interpreting a request to change the end-movement vector of the object, indicating a further status of the object, wherein the further status of the object includes one or more of open, closed, locked, unlocked, safe, not safe, and emergency slow closing when the object is a door when the object is the door, and controlling the set of fluid manipulation emitters to facilitate the second range of shear rates.

18. The head unit system of claim 16, wherein the accessory module further comprises: the first set of processor-executable instructions is further configured to facilitate the first mode of the first range of shear rates, when executed by the processor, to cause the processor to perform one or more of: detecting that a safety threat exists within a first proximity threshold of the object, controlling the motor configured to accelerate the rotational force of the object when the object is a door while the door closes with a first velocity, and indicating a status of the object to include closing of the door with the first velocity; and the second set of processor-executable instructions is further configured to facilitate the second mode of the second range of shear rates, when executed by the processor, to cause the processor to perform one or more of: detecting that the safety threat exists within a second proximity threshold of the object, wherein the second proximity threshold is a greater distance than the first proximity threshold, controlling the motor configured to deaccelerate the rotational force of the object when the object is the door while the door closes with a second velocity, wherein the second velocity is less than the first velocity, and indicating the status of the object to include closing of the door with the second velocity.